A Unified Convergence Theory for Non-Stationary Reinforcement Learning

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Abstract

Non-stationary reinforcement learning lacks a unified convergence theory: variation-budget regret bounds, sliding-window methods, tempo-and-forgetting analyses, two-term decompositions, and causal-RL frameworks each address fragments without composing. We present, to our knowledge, the first *compatible bundle of guarantees* that delivers all three properties — non-stationarity persistence, explicit metric structure on policy space, and on-policy learnability — *jointly*. Each of four components supplies a distinct epistemic property: (i) a two-gap diagnostic separating goal-feasibility from policy-quality routes corrective action; (ii) a point-mass reverse-KL/TV identity under deterministic optimum, $\mathrm{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\mathrm{KL}}(\delta_{a^*} \,\|\, Q)}$ — an *exact* algebraic identity that lies strictly below the Pinsker and Bretagnolle–Huber envelopes at this corner — supplies a coordinate-optimal regret coordinate; (iii) a multi-factor strategic tempo $\mathcal{T}_\Sigma = \min_{(i,j)} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$ with structural survival inequality $\mathcal{T}_\Sigma > \rho_\Sigma / R_\Sigma$ supplies persistence; (iv) closed-loop interventional access under sequential ignorability supplies on-policy learnability via the Pearl causal hierarchy. Composition yields a Best-of-Both-Worlds dynamic regret theorem $\tilde{O}(\min\{V_{\max} N_h \sqrt{(B_T+1)T},\ V_{\max} N_h (V_T+1)^{1/3} T^{2/3}\})$ — automatically adapting between piecewise-stationary segment count B_T and continuous variation budget V_T without prior knowledge of either; the V_T exponent is information-theoretically near-optimal at the deterministic-π^* corner. A structural failure-class theorem identifies the *gain-decay* updates (count-accumulating Bayesian schemes, Robbins–Monro gradient methods) as universally violating the forgetting prerequisite for any positive disturbance rate; finite-gain mechanisms (constant-step stochastic approximation, sliding windows, bounded memory, block restart) face *bidirectional* thresholds instead. The deterministic-optimum scope extends perturbatively to ϵ-stochastic and softmax-regularized policies with $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ corrections (exponentially small with polynomial prefactor). Theory only; we make the ProST reduction rigorous at the steady-state sector-parameter level, lifting it from a single-factor hyperparameter claim to a multi-factor structural-threshold claim.

1 Introduction

Non-stationary reinforcement learning has matured along four largely separate research tracks. Each has produced rigorous results; none has been composed with the others into a single convergence theory.

The four tracks are visible in the recent literature as parallel lineages. **Variation-budget dynamic regret under drift**[1–4] gives sublinear dynamic regret under bounded total variation in reward and transition dynamics. **Two-term regret decompositions**[5–7] split the dynamic regret along an *exploration-vs-adaptation* axis, isolating the cost of confidence-set construction from the cost of tracking a moving optimum. **Tempo and forgetting analyses**[8–12] make policy-update timing or evidence-discount rate an explicit convergence variable. **Causal and interventional access**[13–18] uses causal-graph structure to sharpen sample complexity, but operates in stationary settings only.

The information-theoretic regret literature[19–24] traverses these tracks at a different layer, using entropy, mutual information, information ratio, Pinsker, or Hellinger to bound regret. Notably absent from that literature, as we document below (Appendix F), is an *exact* reverse-KL / total-variation identity at the deterministic-optimum corner: a sharp action-distribution regret coordinate that strictly improves both Pinsker and the Bretagnolle–Huber inequality[25] at this point. The Bretagnolle–Huber inequality is itself absent from the retrieved RL/non-stationary corpus *as an upper-bound regret coordinate* (its use in lower-bound proofs via Le Cam's method[26,27] is well established and not contested here); the *exact identity at its point-mass corner* — which we establish — is *a fortiori* absent.

These four tracks share an evident common ancestor — the dynamic-regret analysis of online MDPs — but no published framework composes them. In particular, no framework combines (a) a regret decomposition along the *goal-feasibility-vs-policy-quality* axis (distinct from the exploration-vs-adaptation decompositions); (b) an exact point-mass reverse-KL/TV regret identity under deterministic optimum, strictly tighter than the Bretagnolle–Huber inequality at this corner; (c) a structural survival inequality threading rate-of-policy-revision against environment-side disturbance; and (d) a closed-loop causal-access argument that makes the regret bound *learnable* on-policy rather than merely provable analytically. The fundamental question remains open:

> *Can we design a unified convergence theory for non-stationary reinforcement learning that handles non-stationarity, has explicit metric structure on policy space, and is learnable from on-policy data?*

In this paper we present, to our knowledge, the first composition that delivers all three properties jointly. Composing four structural elements — a two-gap diagnostic, a point-mass reverse-KL/TV identity, a multi-factor strategic tempo with forgetting prerequisite, and closed-loop interventional access — yields a cumulative dynamic regret theorem with rate $\mathrm{DynReg}(T) \;\le\; \tilde{O}\big(N_h\sqrt{(B_T+1)\,T}\big)$ in the piecewise-stationary regime, where B_T counts piecewise-stationary segment boundaries (kernel-and-reward changes) along the realized trajectory and N_h is the planning horizon. The per-round coordinate of this rate is sharper than both Pinsker and Bretagnolle–Huber at the deterministic-π* corner. The technical anchor is an exact algebraic identity: $\boxed{\mathrm{TV}(\delta_{a^*}, Q) \;=\; 1 - e^{-D_{\mathrm{KL}}(\delta_{a^*}\,\|\,Q)},}$ which holds for any policy Q with $Q(a^*) > 0$ and lies *strictly below* both the Pinsker envelope $\sqrt{D_{\mathrm{KL}}/2}$ and the Bretagnolle–Huber envelope $\sqrt{1-e^{-D_{\mathrm{KL}}}}$ at this corner.

Roadmap Section 2 situates the four-track landscape. Section 3 sets minimal preliminaries. Section 4 presents the four-component composition theorem with narrative unpacking and a defense of why each component is necessary. Section 5 sketches the proof via four key lemmas chained through a block decomposition. Section 6 gives concluding observations and practitioner takeaways. Full proofs and auxiliary mathematical material are in the appendices.

1.1 Contribution

The contribution has the shape — recognized by the NeurIPS Theory Track guidelines as a valid form of originality — of *"a novel combination of existing techniques [where] the reasoning behind this combination is well-articulated."* Each of four components supplies a *distinct epistemic property* the others cannot deliver; together they form a *compatible bundle of guarantees* the field has not previously composed.

Component 1 — corrective-action routing (the two-gap diagnostic, Section 4). A 2×2 split of the regret signal along the *goal-feasibility-vs-policy-quality* axis — the *satisfaction gap* (the goal is unmet under the best available policy) and the *control regret* (the gap between the best policy and the current one) — routes four corrective-action regimes (success, strategy problem, capability limit, both) to revising the policy, the model class or horizon, or the objective. Without this routing, an agent receiving a regret signal cannot distinguish "the goal is unattainable from M_t" from "the current policy is suboptimal." The axis is orthogonal to the exploration-vs-adaptation decompositions of[5–7].

Component 2 — metric layer on policy space (the point-mass identity, Section 4 + Section 5.1). Under deterministic optimum policy $\pi^* = \delta_{a^*}$, the per-round regret coordinate is *exact*: $\mathrm{TV}(\pi^*, Q) = 1 - e^{-D_{\mathrm{KL}}(\pi^* \,\|\, Q)}$. This sits *strictly below* both the Pinsker $\sqrt{D_{\mathrm{KL}}/2}$ and Bretagnolle–Huber $\sqrt{1 - e^{-D_{\mathrm{KL}}}}$ envelopes at this corner — sharper than Pinsker by $7\times$ at $D_{\mathrm{KL}} = 0.01$, with Pinsker becoming vacuous against the trivial $V_{\max}$ envelope for $D_{\mathrm{KL}} > 2$ while the identity saturates at $V_{\max}$ (Appendix E). The identity is *coordinate-optimal among bounds depending only on TV*: any tighter regret bound requires value-landscape information beyond TV. The deterministic-π^* scope is not a hard wall — the identity extends *perturbatively* to ϵ-greedy and softmax-regularized optima with quantified $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ corrections respectively (Section 5.1 + Appendix B.2).

Component 3 — non-stationarity persistence (multi-factor strategic tempo, Section 4 + Section 5.2). A multi-factor strategic tempo composed per revisable policy element of an observation rate ν, an identifiability coefficient ι, and an update gain $1 - \lambda$ yields a structural survival inequality: below threshold $\mathcal{T}_\Sigma > \rho_\Sigma / R_\Sigma$, *no schedule preserves the agent's strategic substate against environment-side disturbance*. All three factors are multiplicatively load-bearing per element, with the bottleneck (per-element minimum) as aggregator — adversarial disturbance concentrates on the slowest-correcting dimension. A structural-class theorem identifies the *gain-decay* class $\mathcal{A}_{\mathrm{decay}}$ — count-accumulating Bayesian updates without forgetting, observation-aggregating schemes without restart, gradient-based methods with vanishing step sizes (Robbins–Monro) — as one in which every agent eventually violates the threshold for any positive disturbance rate; finite-gain mechanisms face *bidirectional* ceilings instead (Appendix C.6). This *lifts* the tempo result of Lee et al. 2023 [8] from a single-factor hyperparameter claim to a multi-factor structural-threshold claim.

Component 4 — on-policy learnability (closed-loop interventional access, Section 4 + Section 5.3). Under sequential ignorability — the agent's information state H_t d-separates a_t from o_{t+1} in the mutilated graph — the closed-loop's data-generating process is *interventional*: by Pearl's do-calculus [28,29], the loop generates Level-2 data by construction. This makes Component 2's regret bound *learnable from on-policy interaction* in regimes with sufficient identifiability, not merely provable in principle. Three regimes partition usable strength: full (Regime A, $\iota \approx 1$), partial (Regime B), observation-only (Regime C), with bias quantified by misidentification probability $1 - p_{\mathrm{id}}$. Coupled-goal architectures violate sequential ignorability and need additional machinery (Section 6).

Composition with non-stationarity (Section 4 + Section 6). The four components compose, via a block decomposition + Cauchy–Schwarz aggregation in the piecewise-stationary case and via the Wei and Luo 2021 [2] MASTER black-box reduction in the continuous-variation case, into a Best-of-Both-Worlds dynamic regret theorem $\mathbb{E}[\mathrm{DynReg}(T)] \leq \tilde{O}\Big(V_{\max}\, N_h \cdot \min\Big\{\sqrt{(B_T+1)\,T},\ (V_T+1)^{1/3} T^{2/3}\Big\}\Big) + V_{\max}\, N_h\, (1 - p_{\mathrm{id}})\, T$, *automatically adapting* to whichever non-stationarity budget — piecewise-stationary segment count B_T or continuous variation V_T — is sharper, without prior knowledge of either. The V_T exponent matches Mao et al. 2021 [3]'s near-optimal rate and is information-theoretically optimal: the lower bound of Besbes et al. 2014 [30] applies at the deterministic-π^* corner. The per-round identity supplies the coordinate that's exact at the corner; it sharpens *constants and per-round form* in both regimes. The bias term is a Regime-A-vanishing value-coordinate misidentification penalty — linear in T but bounded by the trivial $V_{\max} N_h \cdot T$ envelope outside Regime A.

1.2 Scope and limitations

What "convergence" means here We do not claim pointwise policy convergence — under genuine non-stationarity the optimum is itself moving. By *convergence* we mean: per-round regret-coordinate convergence within stationary blocks; cumulative dynamic regret with sublinear / vanishing-average rate when $B_T = o(T)$ (Cesàro tracking $\frac{1}{T}\sum_t (V_t^* - V(\pi_t)) \to 0$); and ultimate boundedness of the strategic mismatch $\|\boldsymbol{\delta}_\Sigma\|$ under the tempo threshold. Pointwise $V(\pi_t) \to V^*$ is structurally unavailable under genuine non-stationarity; the Cesàro tracking statement is the right replacement.

Theory only No experiments. The point-mass identity is mathematically airtight (a two-line direct calculation that specializes the BH family at its deterministic-π^* corner). The empirical grounding for the strategic-tempo claim is provided indirectly through reduction to [8] ProST as a special case (Section 5); at the steady-state sector-parameter level the reduction is rigorous.

Canonical scope The composition theorem holds under deterministic optimum policy, bounded value range, isolated optimum (so $\Delta_{\min} > 0$), and a singular causal trajectory in the sense made precise in Section 3. The deterministic-π^* scope extends *perturbatively* to ϵ-stochastic and softmax-regularized optima (Section 5.1); tied-optimum is a degraded one-sided form (Appendix C.5). Each axis of the canonical scope degrades cleanly outside it (we discuss the failure modes in Section 6).

Comparator regime The cumulative rate is in the piecewise-stationary B_T family, not the continuous-variation V_T family of [3]. Generalizing the per-round-identity route to continuous variation is open (Section 6).

2 Related Work

The four tracks named in Section 1 organize the literature.

Variation-budget dynamic regret [1–4,31] establish dynamic regret bounds under a *variation budget* V_T on cumulative reward and transition variation. [3] achieves the near-optimal continuous-variation rate $\tilde{O}(SA\, V_T^{1/3} H\, T^{2/3})$ via RestartQ-UCB; [1] treats the piecewise-stationary case with restart-on-change. Our cumulative rate $\tilde{O}(N_h \sqrt{(B_T+1)T})$ sits in the piecewise-stationary family — same block-shape as Cheung-style restart-on-change — recovered as a corollary of composition with the per-round identity. The novelty is the *threshold form* of the strategic-tempo inequality (environment regimes where no $\lambda \in (0,1)$ stabilizes the sector-Lyapunov mismatch), which dynamic-regret-optimization analyses do not surface.

Two-term regret decompositions along the exploration-vs-adaptation axis [5–7] decompose the dynamic regret into a *confidence-set construction* term and an *adaptation-under-non-stationarity* term. [32] treats a structurally adjacent question — constraint-region feasibility — using a two-term decomposition along that axis. Same shape, different axis: ours is goal-feasibility-vs-policy-quality (the satisfaction gap distinguishes "the goal is unattainable from M_t" from "the current policy is suboptimal"); theirs is uncertainty-source. Neither subsumes the other: a Long-Fei Li-style exploration term can be present in any of our 2×2 cells, and an adaptation term likewise. The two axes are orthogonal.

Tempo and forgetting [8] (ProST) defines an agent tempo as the schedule of policy update times $\{t_1, \ldots, t_K\}$ and computes the schedule that minimizes a dynamic regret upper bound. [9] (Pausing Policy Learning) refines this with non-zero hold durations. [10–12] establish forgetting / sliding-window analyses for non-stationary linear MDPs and bandits. We *lift* the ProST move along two axes simultaneously: single-factor → multi-factor (per-element $\nu \cdot \iota \cdot (1-\lambda)$ rather than scalar update frequency); and hyperparameter-optimization → structural-survival inequality (the threshold below which no schedule persists, irrespective of which schedule is chosen). ProST recovers as the Beta-Bernoulli special case $|E| = 1, \nu = \iota = 1$.

Causal and interventional access [13–18,33] use causal-graph structure to sharpen sample complexity in RL. Their setting is *stationary*. We compose the closed-loop interventional-access argument with non-stationarity. The connection between Pearl's causal hierarchy [28] and the data-generating process of an agent in a feedback loop [29] is the technical bridge.

Information-theoretic regret (cross-cutting) [19–24,34] traverse the above tracks at a different layer, using entropy / mutual information / information ratio / Pinsker / Hellinger as the divergence backbone. The Russo–Van Roy information ratio uses Pinsker as its Cauchy–Schwarz follow-up; the IDS line and its descendants use mutual-information bounds. The Bretagnolle–Huber inequality [25] — well-known in lower-bound proofs via Le Cam's method [26,27] — does not appear *as an upper-bound regret coordinate* in this corpus, and its exact point-mass corner (our Lemma 5.1) is *a fortiori* absent. We document the retrieval scope and methodology in Appendix F.

Adjacent (satisficing / feasibility) [35,36] use *satisficing* to mean "any policy above acceptance level β is acceptable." Our δ_{sat} is structurally distinct: $V_O^{\min}$ is a property of the objective (set by the application domain), and $\delta_{\text{sat}} > 0$ signals goal-relative *infeasibility* (no policy in Π meets the threshold), not policy-relative satisficing.

Contemporaneous (post-March 2026) [36,37]; cited and distinguished without empirical-comparison demand per the contemporaneous-work cutoff convention.

3 Preliminaries

We adopt strict minimalism: only the notation and assumptions used directly in the main result and the mechanism statements appear here. Convention-hierarchy details (one-step vs. receding-horizon vs. Bellman continuation), the action-gap matching argument, tied-optimum extensions, and the deterministic-trajectory deployment modes are deferred to Appendix C.

Episodic non-stationary MDP A finite-horizon non-stationary MDP $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$ with state space $\mathcal{S}$, finite action space $\mathcal{A}$, time-indexed transition kernels $P_t(\cdot \mid s, a)$ and reward functions $r_t(s, a)$ allowed to vary across rounds t, and finite planning horizon N_h. Standard non-stationary-RL boundedness assumptions[1]: per-step reward in $[0, 1]$.

Variation regimes A variation budget V_T measures continuous variation in transitions and rewards. We work in the *piecewise-stationary* specialization: time partitions into $B_T + 1$ maximal stationary intervals on which (P_t, r_t) is constant, with $B_T := |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$ counting *kernel-stationary-segment boundaries*. Optimum-changes $a_t^* \neq a_{t-1}^*$ can only occur at segment boundaries (since a^* is determined by (P_t, r_t)); the converse is not generally true (a segment boundary need not change a^* — e.g., uniform reward shifts that lift all actions, or kernel shifts that preserve the argmax). Counting kernel-segment boundaries rather than optimum-change events is the right measure for the base learner's per-block stationary-regret guarantee in (A5): UCRL2 / UCBVI / Thompson sampling build empirical confidence sets on (P, r), and within-block kernel drift breaks the per-block $\sqrt{L}$ rate even when a^* is unchanged. In the bandit case with isolated optimum and reward shifts that always cross argmax, B_T reduces to the optimum-change count $|\{t : a_t^* \neq a_{t-1}^*\}|$ — a useful corollary in regime ARG (Section 4 *Comparator regime*) but not the headline framing. The two variation regimes (B_T vs. continuous V_T) are distinct in general; under abrupt-magnitude-δ piecewise-stationarity they coincide up to constants ($V_T \asymp \delta B_T$). Our results are stated for B_T; the continuous-V_T extension is open (Section 6).

Policy distribution The agent's policy at round t is a distribution $Q_t(\cdot \mid s)$ over actions; we write Q when state and round are understood. We adopt the canonical scope: the optimum policy at (M_t, s) is *deterministic*, $\pi^*(\cdot \mid M_t, s) = \delta_{a^*(s)}$ where $a^*(s) := \arg\max_a Q^\pi(s, a)$ is the optimal action under the agent's current model M_t. Deterministic π^* is canonical for finite-MDP RL with isolated optima[27]; the perturbative extension to ϵ-stochastic and softmax-regularized optima (Section 5.1, full derivation Appendix B.2) covers smooth deviations.

Value object Given an objective O, model M_t, policy π, and horizon N_h: $Q_O(M_t, a; \pi_{\text{cont}}, N_h) = \mathbb{E}\Big[V_O(\tau) \;\Big|\; M_t,\, \text{do}(a_t = a),\, a_{t+1:} \sim \pi_{\text{cont}}\Big]$. The $\text{do}(\cdot)$[28] flags intervention on a_t, not conditioning. We adopt the **one-step improvement** convention $\pi_{\text{cont}} = \pi_{\text{current}}$ as default: most conservative, comparable across rounds, fixed-point-free. Receding-horizon and Bellman alternatives are in Appendix C.1.

Bounded value range $V_{\max}(M_t) := \max_a Q_O(M_t, a) - \min_a Q_O(M_t, a)$, finite under bounded reward and finite horizon.

Action gap (isolated optimum) $\Delta(a) := Q_O(a^*) - Q_O(a) \in [0, V_{\max}]$, $\Delta_{\min} := \min_{a \neq a^*} \Delta(a) > 0$ whenever the optimum is isolated.

Strategic regret For policy distribution Q: $R(Q) := Q_O(a^*) - \mathbb{E}_{a \sim Q}[Q_O(a)] = \sum_{a \neq a^*} Q(a) \cdot \Delta(a)$. Three classical forms — $\mathbb{E}_{\pi^*}[V] - \mathbb{E}_Q[V]$, $V(a^*) - \mathbb{E}_Q[V]$, $\mathbb{E}_{\pi^*}[V - V_Q]$ — coincide under deterministic π^*.

Total variation and reverse-KL For finite action measures P, Q on $\mathcal{A}$, $\text{TV}(P, Q) := \frac{1}{2} \sum_a |P(a) - Q(a)|$, $D_{\text{KL}}(P \,\|\, Q) := \sum_a P(a) \log \frac{P(a)}{Q(a)}$ with the convention $0 \log 0 = 0$. Our bounds use the *reverse* KL direction (with $P = \pi^*$); the forward direction $D_{\text{KL}}(Q \,\|\, \pi^*)$ is $+\infty$ whenever Q has off-optimum mass and is therefore vacuous as a regret coordinate (full direction-forcing argument in Appendix C.3).

Singular trajectory The agent operates on a single, non-forkable causal trajectory: a_t causally precedes o_{t+1}, and the model state M_t's sufficiency is defined relative to *this* trajectory rather than to a model-state equivalence class. This is the operational ground for the do-operator above and for the closed-loop interventional-access argument (Section 5.3).

4 Main Result

We present the four components first as motivation, then state the formal composition theorem, then unpack what its conclusions mean.

4.1 The four components

Each component supplies a *distinct epistemic property* that the others cannot deliver. The composition is necessary for the joint properties (rate + persistence + learnability + corrective-action routing), not for the rate alone. The component-by-component failure-mode analysis is in Appendix A.6 below.

Component 1: Two-gap diagnostic — corrective-action routing Let Π denote the policy class available to the agent and let $V_O^{\min}$ denote the threshold trajectory value above which the objective is "met." Define *objective attainability* $A_O(M_t;\, \Pi, N_h) := \sup_{\pi\in\Pi} V_O(M_t, \pi;\, N_h)$, the *satisfaction gap* $\delta_{\text{sat}} := V_O^{\min} - A_O$ (goal-feasibility diagnostic; positive means *unmet* under the best available policy), and the *control regret* $\delta_{\text{regret}} := A_O - V_O(M_t, \pi_{\text{current}};\, N_h) \geq 0$ (policy-quality diagnostic). The 2×2 cross of these two gaps routes corrective action: *Success* (no action), *Strategy problem* (revise the policy), *Capability limit* (revise the model, policy class, or horizon — in that order), *Both* (revise the policy first, then re-evaluate). Without this routing, an agent receiving a regret signal cannot distinguish "the goal is unattainable from M_t" from "the current policy is suboptimal." A single $\delta_{\text{objective}} = V_O^{\min} - V_O(M_t, \pi_{\text{current}}; N_h)$ collapses both regimes into a single number and loses the diagnostic. The two-gap split runs along an axis *orthogonal* to the exploration-vs-adaptation decompositions of[5–7]: theirs isolates uncertainty *sources*; ours isolates *what is wrong*.

Component 2: Point-mass reverse-KL/TV identity — metric layer on policy space Under deterministic $\pi^* = \delta_{a^*}$, the reverse-KL and total variation between π^* and any policy Q are related by the *exact identity* $D_{\text{KL}}(\pi^* \,\|\, Q) = -\log Q(a^*) = -\log\bigl(1-\text{TV}(\pi^*, Q)\bigr) \iff \text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \,\|\, Q)}$. This is a two-line direct calculation under the point-mass assumption (full statement and proof in Section 5.1). It is *not* the Bretagnolle–Huber inequality at equality: substituting it into the BH form $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ gives $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, *strictly* on $D_{\text{KL}} > 0$. The identity sits *strictly below* the BH envelope at this corner. Composed with the textbook TV-regret bound $R(Q) \leq V_{\max}\, \text{TV}(\pi^*, Q)$ and the matching action-gap lower bound, it yields the two-sided characterization $\Delta_{\min}\bigl(1 - e^{-D_{\text{KL}}(\pi^* \,\|\, Q)}\bigr) \leq R(Q) \leq V_{\max}\bigl(1 - e^{-D_{\text{KL}}(\pi^* \,\|\, Q)}\bigr)$, *Lipschitz-equivalent* with constants $\Delta_{\min}/V_{\max}$ and 1. The identity coordinate is *coordinate-optimal among bounds depending only on TV*: any tighter bound on $R(Q)$ requires information beyond TV (the value-landscape spread). Strict improvement over Pinsker is uniform: at $D_{\text{KL}} = 0.01$ the identity bound is $7\times$ tighter; for $D_{\text{KL}} > 2$ Pinsker is vacuous against the trivial $V_{\max}$ envelope while the identity saturates at $V_{\max}$. The deterministic-π^* scope is not a hard wall: the identity extends *perturbatively* to ϵ-greedy and softmax-regularized optima with corrections $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ respectively — exponentially small with polynomial prefactor in the softmax case (Section 5.1).

Component 3: Multi-factor strategic tempo with forgetting prerequisite — non-stationarity persistence Consider a structured policy with $|E|$ revisable elements (in our analysis, edges of an internal causal model the agent maintains over policy components; in Lee et al. 2023[8]'s ProST, sub-policies indexed by a tempo schedule). For each element (i, j), let ν_{ij} denote the per-element observation rate, $\iota_{ij} \in [0, 1]$ the identifiability coefficient (fraction of evidence that genuinely identifies the causal effect rather than reflecting confounded association), and $\eta_{\text{edge},ij}$ the per-element update gain. Within a sector-Lyapunov reduction with diagonal correction (full Model (Σ) in Section 5.2), per-element exponential forgetting at rate λ_{ij} gives steady-state gain $\eta^{\text{ss}}_{\text{edge},ij} \approx 1 - \lambda_{ij}$, and ultimate boundedness of the strategic mismatch $\|\boldsymbol{\delta}_\Sigma\|$ within the modeled dynamics is *sufficient* under the bottleneck threshold $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j)\in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$. All three factors are multiplicatively load-bearing per element (zeroing any collapses the bottleneck; low identifiability cannot be compensated by faster forgetting), and the *bottleneck* (per-element minimum, not aggregate sum) is the right aggregator: a Lyapunov derivation under adversarial disturbance saturates at min, not $\sum$, when the disturbance concentrates on the weakest element. The structural-class theorem identifies the *gain-decay* class $\mathcal{A}_{\text{decay}}$ — count-accumulating Bayesian updates without forgetting (e.g., Beta-Bernoulli with $\eta = 1/(n+1)$), bounded-memory schemes with growing memory, observation-aggregating schemes without restart, *and* gradient-based methods with vanishing step sizes (the Robbins–Monro regime, $\eta_t = c/t^\beta$ for $\beta \in (0,1]$) — as one in which *every* agent eventually violates the threshold for any positive disturbance rate (formal statement in Appendix C.6). Non-accumulating mechanisms (constant-step SA, sliding-window, bounded-memory, block-restart) face *bidirectional* ceilings: the threshold is tight in both directions and the schedule has to be in the right regime, not arbitrarily fast or slow. The familiar single-factor form $(1-\lambda) > \rho_\Sigma / R_\Sigma$ is recovered as a corollary under uniform-Regime-A Beta-Bernoulli normalization.

Component 4: Closed-loop interventional access — on-policy learnability Pearl's causal hierarchy[28] and the causal-hierarchy theorem of[29] establish that Level-2 (interventional) queries $P(Y \mid \mathrm{do}(X))$ are not in general identifiable from Level-1 (associational) data $P(Y \mid X)$. By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes o_{t+1}. Under three conditions on the agent's action mechanism — (C1) *positivity* ($\pi_t(a \mid H_t) \geq p_{\min} > 0$ on the support of the information state H_t), (C2) *sequential ignorability* (the information state blocks unobserved confounding of action selection), and (C3) *known action mechanism* (the behavior policy is known) — conditioning on H_t blocks the action-selection confounding pathway, so the conditional distribution $P(o_{t+1} \mid a_t, H_t)$ coincides with the interventional distribution $P(o_{t+1} \mid \mathrm{do}(a_t), H_t)$. The loop's data-generating process is *interventional* by construction (formal statement in Section 5.3). This makes the regret bound of Component 2 *learnable on-policy* in regimes with sufficient identifiability: the agent's identified optimum $a^*_{\text{ag},t}$ matches the true optimum $\tilde{a}^*_t$ with probability p_{id}, and the per-state, per-step bias from misidentification is bounded by $(1 - p_{\text{id}}(s)) \log(1/q_0)$ under two-point support $Q_t \geq q_0$ (Section 5.4). Three regimes[29] partition usable strength: A (intervention-rich, $\iota \approx 1$), B (partial, $\iota \in (0,1)$), C (observation-only, $\iota \approx 0$).

4.2 The composition theorem

Theorem 4.1 (Composition: joint guarantees with conditional cumulative reduction)**.** *Let $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$ be a non-stationary MDP with bounded reward, finite action space, and for each round t a deterministic optimum policy $\pi^*_t = \delta_{a^*_t}$ with action gap $\Delta_{\min} > 0$. Suppose the agent operates on a singular causal trajectory in the sense of Section 3 and updates its policy with per-element exponential forgetting at rates $\{\lambda_{ij}\}$. Let $K_t(s) := D_{\mathrm{KL}}(\delta_{a^*_t(s)} \,\|\, Q_t(\cdot \mid s))$ be the per-state reverse-KL coordinate (in the bandit case $N_h = 1$ the state argument is suppressed). If*

(A1) Metric extension. *The identity $\mathrm{TV}(\delta_{a^*_t(s)}, Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$ of Lemma 5.1 is read in the extended real sense: when $Q_t(a^*_t(s) \mid s) = 0$, $K_t(s) = +\infty$ and $1 - e^{-K_t(s)} = 1$ recovers the trivial bound $R(Q_t \mid s) \leq V_{\max}$. No pointwise positivity is required; vanilla deterministic UCB / UCBVI / UCRL2 deployed at the optimism-argmax are in scope without modification.*

(A2) Multi-factor forgetting prerequisite. *Within Model (Σ) of Section 5.2, the bottleneck strategic tempo $\mathcal{T}^{\text{bn,ss}}_\Sigma := \min_{(i,j)\in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$.*

(A3) Identifiability regime. *The action mechanism is known and (C1)–(C3) of Section 5.3 hold; the agent operates in Regime A or Regime B with per-state argmax-correctness probability $p_{\text{id}}(s)$, with floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$.*

(A4) Two-gap diagnostic. *The agent applies the satisfaction-gap / control-regret 2×2 to route corrective action.*

(A5) Piecewise-stationary block structure with restart-on-change identity-tight base learner. *The time axis partitions into $B_T + 1$ stationary intervals on which the MDP is stationary and $\pi^*_t(\cdot \mid s) = \delta_{a^*_t(s)}$ is fixed per state. The base learner is* restarted *at each block boundary; within each restarted interval of length L it satisfies $\sum_{t=1}^{L} \mathbb{E}[\overline{\mathrm{TV}}_t] \leq 2c\sqrt{L}$, where $\overline{\mathrm{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} \big[\mathrm{TV}(\pi^*_t(\cdot \mid s_h), Q_t(\cdot \mid s_h))\big]$ is the occupancy-weighted per-round coordinate. (A non-restarting carryover variant is plausible but requires bounding cross-block occupancy shift; see Remark in Section 5.)*

Then:

(i) Per-round regret is two-sided identity-bounded. *At every visited state,* $\Delta_{\min}\big(1 - e^{-K_t(s)}\big) \leq R(Q_t \mid s) \leq V_{\max}\big(1 - e^{-K_t(s)}\big)$.

(ii) Aggregate strategic mismatch is ultimately bounded. *Under (A2), the modeled mismatch* $\|\boldsymbol{\delta}_\Sigma\|$ *is ultimately bounded within* $R^*_\Sigma = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$.

(iii) The KL coordinate is computable with controlled bias. $-\log Q_t(a^*_{\text{ag},t}(s) \mid s)$ *is computable directly from* Q_t. *Per-state bias:* $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$ *under* $Q_t(\cdot \mid s) \geq q_0$ *at the two argmax candidates. Vanishes in Regime A.*

(iv) The 2×2 cell identifies the corrective-action class. $(\delta_{\text{sat}}, \delta_{\text{regret}})$ *routes to revise-policy / revise-model-class-or-horizon / revise-objective.*

(v) Trajectory-level blockwise dynamic regret. *Under (A5),* $\mathbb{E}[\text{DynReg}(T)] := \mathbb{E}\Big[\sum_{t=1}^{T}\big(V_t^{\pi_t^*} - V_t^{Q_t}\big)\Big] \;\leq\; 2c\, V_{\max}\, N_h \sqrt{(B_T+1)\,T} \;+\; V_{\max}\, N_h\, (1 - p_{\text{id}}) \cdot T.$

The proof composes the per-round identity (i) with the simulation lemma[38–41], a block decomposition at optimum-change events, Cauchy–Schwarz across the $B_T + 1$ blocks, and the bias bound from (iii). Full proof in Appendix A; the four key lemmas are surfaced in Section 5.

4.3 Unpacking the conclusions

Theorem 4.1 asserts that the four components compose into a non-stationarity-aware convergence theory with three properties that no existing strand has individually: it *handles non-stationarity* (via (ii)/(A2)), has *explicit metric structure on policy space* (via (i)/Component 2), and is *learnable on-policy* (via (iii)/(A3)). Conclusion (iv) is the connective tissue routing corrective action; conclusion (v) is the cumulative regret rate that follows from composing the per-round metric coordinate with the base-learner stationary rate.

We emphasize four things about this theorem.

The per-round coordinate is exact, not an inequality. Under the canonical scope (deterministic π^*), the per-round identity-bound (i) is $\Delta_{\min}/V_{\max}$-Lipschitz-equivalent to the regret. Both endpoints of the Lipschitz envelope are *achieved* on extremal value landscapes ($\Delta(a) = V_{\max}$ for all $a \neq a^*$ saturates the upper bound; $\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$ saturates the lower). The per-round coordinate is therefore *coordinate-optimal among bounds depending only on TV*: tightening would require value-landscape information beyond TV. This is sharper than Pinsker's $\sqrt{D_{\text{KL}}/2}$ — exact instead of approximate — and sharper than the Bretagnolle–Huber envelope $\sqrt{1 - e^{-D_{\text{KL}}}}$ at this corner since $x < \sqrt{x}$ on $(0, 1)$.

The cumulative rate is in the piecewise-stationary B_T family; the continuous-variation V_T family is reachable via Wei-Luo MASTER black-box wrapping. The lifted rate $\tilde{O}(N_h^2 \sqrt{SA(B_T+1)T})$ that follows under UCRL2/UCBVI base learners (Appendix D) sits in the same block-shape family as the restart-on-change variants of[1] — the route here is through the per-round identity rather than direct optimism. It is *not directly comparable per-form* to the continuous-variation rate $\tilde{O}((SA)^{1/3} V_T^{1/3} H\, T^{2/3})$ of[3]: the parameter (B_T vs. V_T), the T-scaling ($T^{1/2}$ vs. $T^{2/3}$), the S, A-scaling ($\sqrt{SA}$ vs. $(SA)^{1/3}$), and the horizon dependence (N_h^2 vs. N_h) all differ. Under abrupt-magnitude-δ piecewise-stationarity the two regimes coincide up to constants ($V_T \asymp \delta B_T$), and B-CS1's rate is sharper for fixed B_T; outside that, Mao-style continuous-variation analysis is sharper for fixed V_T. The per-round identity route extends to continuous variation via[2]'s MASTER black-box reduction, achieving Best-of-Both-Worlds dynamic regret $\tilde{O}(\min\{\sqrt{(B_T+1)T},\, (V_T+1)^{1/3} T^{2/3}\})$ automatically without prior knowledge of either budget; full development in Section 6 (*On the continuous-variation extension*) and theorem-grade availability via the (A5')-BoBW Remark in Appendix A.

The bias term in (v) is the value-coordinate misidentification penalty. The second term $V_{\max} N_h (1 - p_{\text{id}}) \cdot T$ is the per-round expected value gap between the true and agent-identified optima — bounded by $V_{\max}$ on the misidentification event of probability $1 - p_{\text{id}}$, aggregated over the N_h horizon steps. It vanishes in Regime A ($p_{\text{id}} \to 1$ under (C1)–(C3)); it is linear in T — vacuous as a rate but bounded by the trivial value envelope $V_{\max} N_h \cdot T$ — outside Regime A. In Regime B ($p_{\text{id}} \in (0, 1)$), the dominant rate term is the metric one provided $1 - p_{\text{id}}$ is small enough relative to $\sqrt{(B_T+1)/T}$. In Regime C ($p_{\text{id}} \to 0$) the bound saturates at the trivial envelope; algorithm flags this regime as out-of-scope for on-policy learnability (Appendix D). *Strict assumption-set weakening:* the rate (v) does *not* require Lemma 4's two-point support condition $Q_t(\cdot \mid s) \geq q_0$ — that condition is needed only for conclusion (iii)'s diagnostic computability of the agent's KL coordinate, not for the cumulative-regret rate itself.

The horizon factor is the standard simulation-lemma penalty. The N_h in conclusion (v) is the linear-in-N_h per-round simulation-lemma coefficient — sharper than the N_h^2 of TRPO-style worst-state bounds[42]. The lifted N_h^2 in the UCRL2/UCBVI cumulative rate reflects the additional N_h in the within-block base-learner constant c.

The four components compose into a compatible bundle of guarantees, not a single integrated theorem. The rate (v) goes through (A5) and (i) alone — Component 2 supplies the metric coordinate, (A5) the base-learner regret, and the $\sqrt{(B_T+1)T}$ aggregation falls out. The other components carry the *other* guarantees: Component 1 the corrective-action routing (iv), Component 3 the persistence (ii) (without which K_t may itself drift unboundedly under $\mathcal{A}_{\text{decay}}$ updates), Component 4 the on-policy learnability via (iii) (without which the bound is provable but not realizable in Regime C, where bias saturates and (v) becomes vacuous as a rate). The four together close a story none of the four research lineages they descend from closes individually. Component-by-component failure-mode analysis is in Appendix A.6.

5 Mechanism

This section overviews the proof of Theorem 4.1 via four key lemmas chained through a block decomposition. Each lemma is stated formally; intuitions are given in plain English with the algebra deferred to the appendices. The rate (v) is the composition's load-bearing conclusion; we sketch its derivation in four steps after surfacing the lemmas.

5.1 Key Lemma 1: Point-mass reverse-KL/TV identity

Lemma 5.1 (Point-mass reverse-KL/TV identity)**.** *For deterministic $\pi^* = \delta_{a^*}$ (per visited state, with $a^* = a^*(s)$ and $Q = Q(\cdot \mid s)$) and any policy Q, $D_{\text{KL}}(\pi^* \,\|\, Q) = -\log Q(a^*) = -\log\big(1 - \text{TV}(\pi^*, Q)\big)$, equivalently $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \,\|\, Q)}$. The identity is read in the extended real sense: when $Q(a^*) = 0$ both sides equal 1, with the natural convention $D_{\text{KL}} = +\infty$ and $e^{-\infty} = 0$ — the identity holds for* all *policies Q, including Diracs at suboptimal actions.*

Intuition. The reverse-KL collapses under the point-mass: $D_{\text{KL}}(\delta_{a^*} \,\|\, Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)}$, where the convention $0 \log 0 = 0$ kills every $a \neq a^*$ term, leaving $\log \frac{1}{Q(a^*)} = -\log Q(a^*)$. Total variation collapses just as cleanly: $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2}\sum_a |\delta_{a^*}(a) - Q(a)| = 1 - Q(a^*)$. Combining gives the identity. Substituting into the Bretagnolle–Huber inequality $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \,\|\, Q)}}$ yields $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, *strictly* on $D_{\text{KL}} > 0$. The identity is therefore not BH-at-equality; it is the *exact value* at the deterministic-π* corner, sitting strictly below the BH envelope. Full proof in Appendix B.

Per-round regret bound. Composed with the textbook TV-regret bound $R(Q) \leq V_{\max}\, \text{TV}(\pi^*, Q)$ and the matching action-gap lower bound $R(Q) \geq \Delta_{\min}\, \text{TV}(\pi^*, Q)$, the identity yields the two-sided characterization that gives Theorem 4.1's conclusion (i). The bound is *coordinate-optimal among bounds depending only on TV*: any tighter bound on $R(Q)$ requires value-landscape information beyond TV.

Perturbative extension. The deterministic-π* scope is not a hard wall. For ϵ-greedy π^*_ϵ with Q satisfying full-support $Q(a) \geq q_0$, $\text{TV}(\pi^*_\epsilon, Q) = 1 - e^{-D_{\text{KL}}(\pi^*_\epsilon \,\|\, Q)} + O(\epsilon \log(1/\epsilon))$ uniformly in Q; the proof is a triangle-inequality + expansion argument given in Appendix B.2. For softmax-regularized $\pi^*_\tau \propto \exp(Q_O/\tau)$, the off-optimum mass concentrates as $O(\exp(-\Delta_{\min}/\tau))$ and the correction is $O(\tau^{-1}\exp(-\Delta_{\min}/\tau))$ — exponentially small with polynomial prefactor. The two-sided regret bound transfers with the same correction order. Outside the perturbative regime — genuinely high-entropy optima, tied-optimum, hard-exploration — the BH inequality becomes the relevant general bound; Pinsker is the textbook fallback.

5.2 Key Lemma 2: Forgetting prerequisite

The strategic mismatch state $\boldsymbol{\delta}_\Sigma = (\delta_{ij})_{(i,j)\in E}$ in the Euclidean norm $\|\boldsymbol{\delta}_\Sigma\|^2 = \sum_{(i,j)} \delta_{ij}^2$ evolves in discrete time as $\mathbb{E}[\Delta\delta_{ij} \mid \delta_{ij}] = -\alpha_{ij}\delta_{ij} + w_{ij}, \qquad |w_{ij}| \le \rho_\Sigma/|E|^{1/2}$, where $\alpha_{ij} \ge 0$ is a per-element correction rate (the strategic sector parameter), w_{ij} is a bounded disturbance, and the disturbance budget satisfies $\sum_{(i,j)} w_{ij}^2 \le \rho_\Sigma^2$. The strategic reserve R_Σ is the operative norm-radius beyond which the agent loses tracking. Beta-Bernoulli edge updates instantiate this *Model (Σ)* approximately; gradient-based decaying-step updates instantiate it under standard Robbins–Monro conditions; we will refer to it as Model (Σ) throughout.

Lemma 5.2 (Multi-factor forgetting prerequisite)**.** *Within Model (Σ) with diagonal per-element correction rates $\alpha_{ij} = \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij}$ (per-element observation rate $\nu_{ij} \in [0,1]$, regime-adjusted identifiability $\iota_{ij} \in [0,1]$, edge-update gain $\eta_{\text{edge},ij} > 0$), under per-element exponential forgetting at rate $\lambda_{ij} \in (0,1)$ giving steady-state $\eta^{\text{ss}}_{\text{edge},ij} \approx 1 - \lambda_{ij}$, ultimate boundedness of $\|\boldsymbol{\delta}_\Sigma\|$ within $R^*_\Sigma = \rho_\Sigma/\mathcal{T}^{\text{bn,ss}}_\Sigma$ is* sufficient *under $\mathcal{T}^{\text{bn,ss}}_\Sigma := \min_{(i,j)\in E}\big(\nu_{ij} \cdot \iota_{ij} \cdot (1-\lambda_{ij})\big) > \rho_\Sigma/R_\Sigma$. The threshold is* sharp inside the diagonal sector model*: when the inequality reverses, an adversarial disturbance concentrating on the bottleneck element drives $\|\boldsymbol{\delta}_\Sigma\|$ beyond R_Σ within the modeled dynamics.*

Intuition (why bottleneck, not sum). The Lyapunov sector product is $\boldsymbol{\delta}_\Sigma^\top \mathbf{F}(\boldsymbol{\delta}_\Sigma) = \sum_{(i,j)} \alpha^{\text{ss}}_{ij}\delta^2_{ij}$. Bounding this from below uniformly in $\boldsymbol{\delta}_\Sigma$ requires the largest constant c such that $\sum_{(i,j)} \alpha_{ij}\delta^2_{ij} \ge c \cdot \|\boldsymbol{\delta}_\Sigma\|^2$ holds for all $\boldsymbol{\delta}_\Sigma$. The largest such c is $\min_{(i,j)} \alpha_{ij}$, *not* $\sum_{(i,j)} \alpha_{ij}$ — adversarial concentration of the disturbance on the *weakest* element saturates the bound at the bottleneck. The same effect makes per-dimension persistence conditions sharper than scalar conditions in adaptive control[43]: scalar tempo can overstate the available margin because the worst-case disturbance concentrates on the slowest-correcting dimension. Ultimate boundedness then follows from Khalil's classical sector-Lyapunov template applied to the strategic substate. Full proof in Appendix B; the structural-class theorem on $\mathcal{A}_{\text{decay}}$ (gain-decay updates universally violate the threshold) is in Appendix C.6.

ProST as a sector-level reduction. The block-restart structure of ProST[8] recovers as the *impulsive limit* of Model (Σ): between updates the mismatch evolves under disturbance alone, and each update contracts the mismatch by per-update gain γ. The Hespanha–Liberzon–Teel reverse-ADT framework[44] gives the threshold (full development in Appendix B.4); under uniform schedules this recovers ProST's K/T form with the impulse gain γ made explicit. The reframe lifts ProST's *tempo-as-hyperparameter* framing into *tempo-as-stability-margin* — the longest inter-update gap, not the average, sets the threshold, exposing the tradeoff between update frequency and per-update impulse strength at the sector-Lyapunov level.

5.3 Key Lemma 3: Closed-loop interventional access

Lemma 5.3 (Loop generates interventional samples)**.** *Let H_t denote the agent's information state at round t (model state M_t, history of observations and actions). Under (C1)* positivity *— $\pi_t(a \mid H_t) \ge p_{\min} > 0$ on the support of H_t for the actions of interest; (C2)* sequential ignorability *— H_t d-separates a_t from o_{t+1} in the mutilated graph $G_{\overline{a_t}}$ (equivalently, $a_t \perp\!\!\!\perp Y^{(a_t)} \mid H_t$ in potential-outcome notation); (C3)* known action mechanism *— the behavior policy $\pi_t(a \mid H_t)$ is known: executed actions are interventions relative to the environment's transition kernel, and the loop generates samples from $P(o_{t+1} \mid \mathrm{do}(a_t = a), H_t)$ — i.e., Pearl Level-2 interventional kernels conditional on the decision context. The substantive content for the identification claim is (C2); under the singular-trajectory + agent-as-policy architecture of Section 3, (C1) reduces to realized-action positivity (automatic) and (C3) to agent-policy-queryability (architectural). (C1) and (C3) are stated explicitly for forward-compatibility with off-policy / counterfactual extensions.*

Intuition. By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes o_{t+1}. Under (C1)–(C3), conditioning on the information state H_t blocks the action-selection confounding pathway, so the conditional distribution $P(o_{t+1} \mid a_t, H_t)$ coincides with the interventional distribution $P(o_{t+1} \mid \mathrm{do}(a_t), H_t)$ on the support where $\pi_t(a_t \mid H_t) > 0$. The architecture of the agent (Q-learning, transformer, etc.) does not enter: the loop generates conditional-Level-2 samples whenever (C1)–(C3) hold, irrespective of how the agent updates internally. Whether the agent *exploits* this data for Level-2 reasoning is a separate algorithmic question.

Three regimes of identifiability. Three regimes [29] partition usable strength of identification: - **Regime A (intervention-rich, $\iota \approx 1$).** Software tests, controlled labs. do-effects identified cleanly; bound realizable on-policy. - **Regime B (partial, $\iota \in (0,1)$).** Mixed observation-intervention. Bound holds for the model the agent identifies; bias $\propto 1-\iota$. - **Regime C (observation-only, $\iota \approx 0$).** Passive monitoring. Bound provable analytically but not realizable on-policy; observer-on-sub-agent extensions can recover identifiability in subcases (Appendix C.8).

Action-generated data is Level 2 in *character*, but a clean estimate of $P(o \mid \mathrm{do}(a), H_t)$ requires overcoming four typical obstacles: coverage (diverse actions tried), within-step confounding, delay (consequences past $t+1$), and partial observability. We honor this distinction. Components 2 and 4 are jointly load-bearing: the identity of Section 5.1 gives the analytic regret-KL relationship; the loop-Level-2 access here gives the data substrate from which π^* — and the bound's RHS — can be empirically estimated under sufficient identifiability. Neither alone makes the bound learnable.

5.4 Key Lemma 4: Bias bound under partial identifiability

Lemma 5.4 (Bias bound)**.** *Assume $Q_t(a \mid s) \geq q_0 > 0$ for both $a = a^*_{\mathrm{ag},t}(s)$ (the agent's identified optimum) and $a = \tilde{a}^*_t(s)$ (the true optimum). Let $\hat{D}_t(s) := -\log Q_t(a^*_{\mathrm{ag},t}(s) \mid s)$ and $D^{\mathrm{true}}_t(s) := -\log Q_t(\tilde{a}^*_t(s) \mid s)$. Then $\left|\hat{D}_t(s) - D^{\mathrm{true}}_t(s)\right| \leq \mathbf{1}[a^*_{\mathrm{ag},t}(s) \neq \tilde{a}^*_t(s)] \cdot \log(1/q_0)$, and taking expectations: $\mathbb{E}[|\hat{D}_t(s) - D^{\mathrm{true}}_t(s)|] \leq (1 - p_{\mathrm{id}}(s)) \log(1/q_0)$, where $p_{\mathrm{id}}(s) := \Pr[a^*_{\mathrm{ag},t}(s) = \tilde{a}^*_t(s)]$.*

Intuition. The argument is an indicator decomposition. On the matching event, the difference is identically zero. On the misidentification event, $\log Q_t(a^*_{\mathrm{ag},t}(s) \mid s)$ and $\log Q_t(\tilde{a}^*_t(s) \mid s)$ both lie in $[\log q_0, 0]$, so their absolute difference is at most $\log(1/q_0)$. The expectation bound follows from $\Pr[\text{misidentification event}] = 1 - p_{\mathrm{id}}(s)$. Bias scales *linearly* in misidentification mass — vanishes in Regime A ($p_{\mathrm{id}} \to 1$), controlled in Regime B, saturates at $\log(1/q_0)$ in Regime C. The argmax-correctness probability p_{id} relates to the per-edge identifiability ι via the policy DAG structure; in the simple-bandit case $p_{\mathrm{id}} = \iota$. Note: the support condition is *two-point* (only at the two argmax candidates), strictly weaker than the *full-support* condition $Q(a) \geq q_0$ for all a that the perturbative identity (Section 5.1) requires — F.1 applies whenever the two candidates are positively-supported regardless of Q_t's mass on other actions. Full proof in Appendix B.

5.5 Combining the lemmas: cumulative regret in four steps

We sketch the proof of conclusion (v) of Theorem 4.1 — the cumulative trajectory-level dynamic regret rate — by chaining the four key lemmas through a block decomposition.

Step 1 — Per-round bound via the simulation lemma. For finite-horizon N_h non-stationary MDPs with bounded reward, the standard performance-difference / simulation lemma [38–41] gives, for any two policies π^*_t and Q_t and initial state s_0, $V^{\pi^*_t}_t(s_0) - V^{Q_t}_t(s_0) \leq V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d^{Q_t}_h}\left[\mathrm{TV}(\pi^*_t(\cdot \mid s_h), Q_t(\cdot \mid s_h))\right] = V_{\max} \cdot N_h \cdot \overline{\mathrm{TV}}_t$, where $d^{Q_t}_h$ is the round-t learner-induced state distribution and $\overline{\mathrm{TV}}_t$ is the occupancy-weighted TV defined in (A5). By Key Lemma 1 applied per state, $\mathrm{TV}(\pi^*_t(\cdot \mid s), Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$ — strictly sharper than Pinsker — preserved underneath the occupancy aggregation. The horizon factor N_h here is the linear-in-N_h simulation-lemma penalty, sharper than the N_h^2 of TRPO-style worst-state bounds [42].

Step 2 — Block decomposition at piecewise-stationary segment boundaries. Partition $[1, T]$ at (P, r)-change events into $0 = \tau_0 < \tau_1 < \cdots < \tau_{B_T} \leq \tau_{B_T+1} = T$ (the kernel-stationary-segment boundaries of Section 3). Within each block $[\tau_i, \tau_{i+1})$ of length Δ_i the MDP is stationary by construction. (A5) — restart-on-change base learner — gives within-block guarantee $\sum_{t \in \mathrm{block}_i} \mathbb{E}[\overline{\mathrm{TV}}_t] \leq 2c\sqrt{\Delta_i}$.

Step 3 — Cauchy–Schwarz across blocks. Across the $B_T + 1$ blocks, $\sum_i \sqrt{\Delta_i} \leq \sqrt{(B_T+1)T}$ by Cauchy–Schwarz. Multiplying by $V_{\max} N_h$ closes the rate term: $\sum_t \mathbb{E}[V_{\max} N_h \overline{\mathrm{TV}}_t] \leq 2cV_{\max} N_h \sqrt{(B_T+1)T}$.

Step 4 — Misidentification penalty via simulation lemma at the identified-vs-true-optimum gap. Steps 1–3 give the rate term against the agent's *identified* optimum policy $\pi^*_{\mathrm{ag},t} := \delta_{a^*_{\mathrm{ag},t}}$ (which is what (A5)'s base-learner guarantee delivers); the dynamic regret in (v) compares to the *true* optimum $\tilde{\pi}^*_t := \delta_{\tilde{a}^*_t}$. Apply the simulation lemma to these two point-mass policies: each per-step bracket reduces to $Q_h^{\tilde{\pi}^*}(s_h, \tilde{a}^*) - Q_h^{\tilde{\pi}^*}(s_h, a^*_{\text{ag}}) \in [0, V_{\max}] \cdot \mathbf{1}[\text{misid at } s_h]$ (zero on the matching event by optimality of $\tilde{a}^*$, bounded by $V_{\max}$ on the misidentification event). Aggregated over the N_h horizon steps with floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$, the per-round misidentification penalty is at most $V_{\max} N_h (1 - p_{\text{id}})$, contributing $V_{\max} N_h (1 - p_{\text{id}}) \cdot T$ to cumulative regret — *in value coordinates* throughout, no detour through the KL coordinate. (Key Lemma 4's KL-readout bound is the right tool for conclusion (iii)'s diagnostic computability; for the value-coordinate penalty here, indicator decomposition is direct and tighter — and the support condition $Q_t \geq q_0$ is *not* required for this term, only for conclusion (iii).) Vanishes in Regime A.

Combining Steps 1–4 gives conclusion (v): $\mathbb{E}[\text{DynReg}(T)] \leq 2cV_{\max} N_h \sqrt{(B_T+1)T} + V_{\max} N_h (1 - p_{\text{id}}) \cdot T$. The Cesàro tracking corollary $\frac{1}{T}\sum_t (V_t^{\tilde{\pi}_t^*} - V_t^{Q_t}) = O(V_{\max} N_h \sqrt{(B_T+1)/T}) \to 0$ (when $B_T = o(T)$ and $p_{\text{id}} \to 1$, i.e., Regime A) follows by dividing through by T.

Bandit case sharpening In the bandit special case ($N_h = 1$), Thompson sampling and UCB satisfy (A5) with the *stronger logarithmic rate* $\mathbb{E}[1 - e^{-K_t}] = O(\log t / (t\,\Delta^2_{\min}))$ per stationary block (Appendix D) — by Lattimore and Szepesvári 2020[27] Theorem 7.1, $\mathbb{E}[N_a(L)] = O(\log L/\Delta_a^2)$ per suboptimal arm, aggregating to the per-round form. Cumulative regret $O(V_{\max}(B_T+1)\log(T/(B_T+1))/\Delta^2_{\min})$ — sharper than the worst-case $\sqrt{(B_T+1)T}$ when $\Delta_{\min}$ is bounded away from zero. The framework's $V_{\max} \cdot$ TV chain is structurally one factor of $\Delta_{\min}$ looser than direct gap-aware UCB analysis (which gives $O(\log T/\Delta_{\min})$ cumulative by weighting each suboptimal pull by its own $\Delta(a)$ rather than by $V_{\max}$); this is a feature of the unification — the framework's contribution is composition, not a sharper bandit constant.

Carryover variant of (A5) If the base learner does *not* restart at block boundaries — e.g., a single UCRL2 instance run across the whole horizon — the within-block $\sqrt{L}$ guarantee can fail at switch events because $d_h^{Q_t}$ at the start of block $i+1$ reflects the previous block's policy rather than a fresh exploration distribution. Bounding the cross-block occupancy shift requires either (a) detecting switches and locally re-exploring (which recovers a restart-style analysis), or (b) showing the base learner's regret is robust to mis-calibrated state distributions. Sliding-window or adaptive-window variants[1,2] can be analyzed this way under additional structural assumptions. Theorem 4.1(v) is stated for the restart case; the carryover case is plausible but requires bounding the occupancy shift, deferred as future work.

Detailed full proof of (v) in Appendix A.

6 Conclusion

We assemble four components — the two-gap diagnostic, the point-mass reverse-KL/TV identity, the multi-factor strategic-tempo forgetting prerequisite, and closed-loop interventional access — into a unified non-stationary RL convergence theory. The composition delivers three properties no published strand has individually: it *handles non-stationarity* (via the strategic-tempo persistence inequality), has *explicit metric structure on policy space* (via the point-mass identity), and is *learnable on-policy* (via closed-loop interventional access). The cumulative dynamic regret rate $\tilde{O}(N_h\sqrt{(B_T+1)T})$ matches the piecewise-stationary B_T family of[1] as a corollary; the per-round identity coordinate is *exact* under deterministic π^* and strictly sharper than Pinsker and Bretagnolle–Huber at this corner. The point-mass identity is the technical anchor — absent, to our knowledge, from the prior RL/non-stationary corpus as an upper-bound regret coordinate.

We close with concluding observations and practitioner takeaways.

6.1 Concluding observations

On the optimal dependencies on N_h, S, A The lifted UCRL2/UCBVI rate $\tilde{O}(N_h^2\sqrt{SA(B_T+1)T})$ has linear-in-N_h per-round penalty (the simulation lemma) plus an additional N_h from the within-block constant c. Bernstein-type concentration in the base learner could potentially shave one $\sqrt{N_h}$ factor, by analogy with[41] for tabular non-stationary MDPs. We believe the $\sqrt{N_h}$ gap between this and the corresponding tabular non-stationary lower bound is expected because the base-learner concentration used here is intrinsically Hoeffding-type.

On the continuous-variation extension The current rate is in the piecewise-stationary B_T family (kernel-stationary-segment count, Section 3). The same per-round identity route extends to the continuous-variation V_T regime via the black-box reduction of[2] (MASTER), which wraps any stationary base learner with $\tilde{O}(\sqrt{L})$ regret into a non-stationary algorithm achieving $\tilde{O}(\min\{\sqrt{(B_T+1)T},\ (V_T+1)^{1/3}T^{2/3}\})$ Best-of-Both-Worlds dynamic regret without prior knowledge of either variation budget. Our framework's identity-routed (A5)-compatible learner — achieving $\sum_{t\in\text{block}} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{L}$ in TV coordinates per stationary block — satisfies MASTER's base-learner condition (Wei-Luo Assumption 1, with $C(t) = c\,t^{1/2}$, $p = 1/2$ at the boundary case explicitly admitted). The wrapping is mechanical: the per-round identity (Steps 1–2 of Appendix A) is stationarity-agnostic and survives the substitution of block-Cauchy–Schwarz aggregation by adaptive-window MASTER aggregation; the misidentification penalty (Step 4) is round-additive and rides through. Theorem-grade availability in Appendix A (Remark on (A5')-BoBW). The achieved $V_T^{1/3}T^{2/3}$ exponent matches[3]'s near-optimal RestartQ-UCB rate $\tilde{O}((SA)^{1/3}\Delta^{1/3}HT^{2/3})$ and is information-theoretically near-optimal: the[30] lower bound $\Omega(K^{1/3}V_T^{1/3}T^{2/3})$ for non-stationary stochastic bandits applies *also at the deterministic-π corner** (their hard instance is a Bernoulli MAB with dynamic oracle playing the per-round argmax — deterministic-per-round optimum), so the per-round identity sharpens *constants and per-round form*, not the V_T exponent. The two regimes are *not* structurally distinct: they share the same per-round coordinate $1-e^{-K_t}$ and differ only in *aggregation strategy* (block-Cauchy–Schwarz across B_T+1 stationary intervals vs. adaptive-window MASTER over $\log T$ parallel base-learner instances). MASTER also gives the parameter-free property — automatic adaptation to whichever budget is sharper without prior knowledge of B_T or V_T — that Mao 2021's basic RestartQ-UCB requires[3]'s Double-Restart variant for. Two refinements remain open: an *agent-experienced* continuous-variation budget $V_T^{\text{eff},\Sigma}$ paralleling the $B_T^{\text{eff},\Sigma}$ direction below, and an *identity-coordinate* variation budget $V_T^{(K)} := \sum_t |K_t - K_{t-1}|$ that lives on the action-distribution coordinate and could give sharper-than-V_T rates in absorbing regimes where the strategic substate damps environment-side drift.

On agent-experienced block count The B_T of the rate (v) is an *environment-side* quantity — every kernel-and-reward change counts. The strategic-tempo guarantee (ii) suggests an *agent-experienced* block count $B_T^{\text{eff},\Sigma}$ — the number of times $\|\boldsymbol{\delta}_\Sigma\|$ exits its ultimate-bounded region — could be strictly smaller than B_T in regimes where per-element forgetting absorbs within-segment drift. Whether (v) sharpens to $\sqrt{(B_T^{\text{eff},\Sigma}+1)\,T}$ requires algorithmic design coupling the base learner's confidence-set widening with the strategic substate $\boldsymbol{\delta}_\Sigma$, deferred to follow-up. The result would explain why agents in high-V_T slow-drift environments often outperform B_T-pessimistic predictions: what matters is the *agent's effective* variation budget, not the environment's.

On joint uniqueness of the four-component composition The metric layer is uniquely reverse-KL up to positive scaling under chain-rule additivity (Theorem C.3 in Appendix C.10) — any theory satisfying explicit metric structure plus chain-rule must use reverse-KL. We do *not* claim joint uniqueness across all three properties (non-stationarity / metric / learnable). Alternative correction architectures (non-sector-bounded revisions; non-diagonal correction) and alternative interventional-access taxonomies exist in principle, and a joint-uniqueness theorem would require further axioms (singular-trajectory + sector-boundedness + identifiability conditions) — open.

On coupled-goal architectures (C2)'s sequential-ignorability — that H_t d-separates a_t from o_{t+1} in the mutilated graph $G_{\overline{a_t}}$ — presupposes architectural separation (conditional independence) between the model state M_t (in H_t) and the goal state. Goal-conditioned LLM policies and similar coupled-goal architectures violate this separation: the goal influences both action selection (via the policy) and the environment's response to the action (via shared latent context such as user intent, dialogue history, or task framing), so H_t does not block the goal-mediated path from a_t to o_{t+1} — (C2) fails and the loop's data-generating process is no longer interventional in our sense. The conditional-independence relaxation results of[45] are a starting point for handling such cases. We treat this as an interesting boundary case that the current framework does not directly cover.

On strict tightness of the strategic-tempo threshold The bottleneck threshold $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma/R_\Sigma$ is *sufficient and sharp inside the diagonal sector model* — when reversed, an adversarial disturbance concentrating on the bottleneck element drives the modeled mismatch beyond R_Σ. The theorem is silent about correction architectures *outside* the sector-Lyapunov reduction (e.g., non-diagonal correction, alternative non-sector-bounded revisions, or stabilization produced by mechanisms other than the modeled α_Σ-correction). Whether the threshold extends to those architectures is a useful open question in the adaptive-control–RL bridge.

6.2 Practitioner takeaways

For a practitioner seeking to apply the four-component theory:

1. **The point-mass identity at the deterministic-π* corner is a corner-case sharpening of Bretagnolle–Huber that hasn't been deployed in RL regret upper bounds.** Where deterministic π* applies — most tabular finite-MDP regret analyses with isolated optima — the identity yields *exact* regret-vs-KL Lipschitz equivalence rather than the loose Pinsker / BH inequalities. The per-round coordinate $1 - e^{-D_{\mathrm{KL}}}$ is computable directly from Q and is strictly sharper than $\sqrt{D_{\mathrm{KL}}/2}$ at every $D_{\mathrm{KL}} > 0$.
2. **The multi-factor strategic tempo lifts ProST's tempo-as-hyperparameter framing into a structural survival inequality.** Three factors (observation rate ν, identifiability ι, discount rate $1 - \lambda$) are independently load-bearing per element. A practitioner can use the aggregate necessary-condition $\mathcal{T}_\Sigma^{\mathrm{agg,ss}} > |E| \cdot \rho_\Sigma / R_\Sigma$ as a fail-fast pre-check and the variation-budget estimation literature[1,2,37] to estimate the environment-side ρ_Σ / R_Σ from observed reward and transition variation.
3. **Closed-loop interventional access makes regret bounds learnable on-policy under sufficient identifiability.** Regime A (intervention-rich, $\iota \approx 1$) makes the bound directly realizable; Regime B (partial) carries quantified bias linear in $1 - p_{\mathrm{id}}$; Regime C (observation-only) requires observer-on-sub-agent extensions (Appendix C.8) or flags the regime as out-of-scope. The 2×2 diagnostic routes corrective action when regret is non-zero — distinguishing "policy revision" from "model / policy class / horizon revision" from "objective revision."

We hope this work contributes a unifying lens for the four-track non-stationary RL landscape, and that the point-mass identity is a useful tool in subsequent regret analyses where deterministic π* is the canonical scope.

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A Proof of the Composition Theorem

This appendix gives the full proof of Theorem 4.1. The proof composes the four key lemmas of Section 5 (full proofs of those in Appendix B) with a block decomposition at optimum-change events plus Cauchy–Schwarz across blocks.

Notation Throughout this section: $K_t(s) := D_{\mathrm{KL}}(\delta_{a_t^*(s)} \,\|\, Q_t(\cdot \mid s))$ is the per-state reverse-KL coordinate; $d_h^{Q_t}(\cdot \mid s_0)$ is the round-t learner-induced state distribution at horizon step h starting from s_0; $\overline{\mathrm{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}}[\mathrm{TV}(\pi_t^*(\cdot \mid s_h), Q_t(\cdot \mid s_h))]$ is the occupancy-weighted per-round coordinate of (A5); $B_T = |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$ counts piecewise-stationary segment boundaries (per Section 3); the partition is $0 = \tau_0 < \tau_1 < \cdots < \tau_{B_T} \leq \tau_{B_T+1} = T$ with block i length $\Delta_i := \tau_{i+1} - \tau_i$.

A.1 Proof of (i) — Per-round identity-bounded regret

For deterministic $\pi_t^* = \delta_{a_t^*(s)}$ and any $Q_t(\cdot \mid s)$, Key Lemma 1 (Lemma 5.1, full proof in Appendix B.1) gives the exact identity $\mathrm{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$ in the extended real sense (when $Q_t(a_t^*(s) \mid s) = 0$, $K_t(s) = +\infty$ and both sides equal 1, recovering the trivial bound). Composed with the per-state TV-regret bounds (full proofs in Appendix B.1 and Appendix C.4): $\Delta_{\min} \cdot \mathrm{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) \;\leq\; R(Q_t \mid s) \;\leq\; V_{\max} \cdot \mathrm{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s))$, substituting the identity yields conclusion (i): $\Delta_{\min}(1 - e^{-K_t(s)}) \leq R(Q_t \mid s) \leq V_{\max}(1 - e^{-K_t(s)})$. $\square$

A.2 Proof of (ii) — Aggregate mismatch ultimate boundedness

Under (A2), Key Lemma 2 (Lemma 5.2, full proof in Appendix B.3) applied to the diagonal sector model with $\alpha_{ij}^{\mathrm{ss}} = \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$ gives ultimate boundedness of $\|\boldsymbol{\delta}_\Sigma\|$ within $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\mathrm{bn,ss}}$. $\square$

A.3 Proof of (iii) — KL coordinate bias bound

Key Lemma 4 (Lemma 5.4, full proof in Appendix B.6) gives the per-state bias bound directly, under (A3)'s identifiability assumption (which gives $p_{\mathrm{id}}(s)$ for each s) and the two-point support condition $Q_t(\cdot \mid s) \geq q_0$ at $a_{\mathrm{ag},t}^*(s)$ and $\tilde{a}_t^*(s)$. $\square$

A.4 Proof of (iv) — 2×2 corrective-action routing

(A4) states the agent applies the satisfaction-gap / control-regret 2×2 (Section 4.1 Component 1). Conclusion (iv) is a direct restatement of the 2×2 cell partition. The decomposition is convention-dependent (the values of δ_{sat} and δ_{regret} depend on the continuation convention; see Appendix C.1 for monotonicity across C1/C2/C3); the 2×2 *structure* is preserved across all three conventions. $\square$

A.5 Proof of (v) — Trajectory-level cumulative regret

We chain Key Lemmas 1 and 4 with the simulation lemma and a Cauchy–Schwarz block argument. The proof is in four steps; the high-level sketch is in Section 5.5.

Step 1: Simulation / performance-difference lemma. For any two policies π and π', any state s, and finite-horizon non-discounted MDP with horizon N_h, the standard performance-difference lemma[38–41] gives $V_t^{\pi'}(s) - V_t^{\pi}(s) = \mathbb{E}\Big[\sum_{h=0}^{N_h-1}\Big(\mathbb{E}_{a' \sim \pi'(\cdot \mid s_h)}[Q_h^{\pi'}(s_h, a')] - Q_h^{\pi'}(s_h, a_h)\Big) \;\Big|\; s_0 = s,\ a_h \sim \pi\Big]$. Each per-step bracket is bounded by value range times TV: $\big|\mathbb{E}_{a' \sim \pi'}[Q_h^{\pi'}] - \mathbb{E}_{a \sim \pi}[Q_h^{\pi'}]\big| \;\leq\; V_{\max} \cdot \mathrm{TV}(\pi'(\cdot \mid$

$s_h), \pi(\cdot \mid s_h))$. Applied at round t with $\pi' = \pi^*_t$, $\pi = Q_t$: $V_t^{\pi^*_t}(s_0) - V_t^{Q_t}(s_0) \leq V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}(\cdot|s_0)}\big[\mathrm{TV}(\pi^*_t(\cdot \mid s_h), Q_t(\cdot \mid s_h))\big] = V_{\max} \cdot N_h \cdot \overline{\mathrm{TV}}_t$. The horizon factor N_h here is the *linear-in-N_h* simulation-lemma penalty — sharper than the N_h^2 of TRPO-style worst-state bounds [42], which bound max-state TV rather than occupancy-weighted TV.

Step 2: Per-state TV identity along the trajectory. By the deterministic-per-state optimum scope of Theorem 4.1 ($\pi^*_t(\cdot \mid s) = \delta_{a^*_t(s)}$), at every visited state Key Lemma 1 gives $\mathrm{TV}(\pi^*_t(\cdot \mid s), Q_t(\cdot \mid s)) = 1 - Q_t(a^*_t(s) \mid s) = 1 - e^{-K_t(s)}$. Summing over the trajectory and taking expectations: $\overline{\mathrm{TV}}_t = \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}}[1 - e^{-K_t(s_h)}]$. The per-state KL/TV identity is preserved underneath the occupancy aggregation; the per-round coordinate is sharper than Pinsker/BH at every visited state.

Step 3: Block-sum + Cauchy–Schwarz across blocks. Within each block $[\tau_i, \tau_{i+1})$ of length Δ_i, (A5)'s restart-on-change identity-tight base learner gives $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\mathrm{TV}}_t] \leq 2c\sqrt{\Delta_i}$. Summing over the $B_T + 1$ blocks: $\sum_{t=1}^{T} \mathbb{E}[\overline{\mathrm{TV}}_t] = \sum_{i=0}^{B_T} \sum_{t \in \text{block}_i} \mathbb{E}[\overline{\mathrm{TV}}_t] \leq 2c \sum_{i=0}^{B_T} \sqrt{\Delta_i}$. Cauchy–Schwarz gives $\sum_{i=0}^{B_T} \sqrt{\Delta_i} \leq \sqrt{(B_T+1) \cdot \sum_i \Delta_i} = \sqrt{(B_T+1)T}$. Multiplying by $V_{\max} N_h$: $\sum_{t=1}^{T} \mathbb{E}[V_{\max} N_h \overline{\mathrm{TV}}_t] \leq 2c V_{\max} N_h \sqrt{(B_T+1)T}$. This is the rate term.

Step 4: Misidentification penalty via simulation lemma at the identified-vs-true-optimum gap. Steps 1–3 implicitly compared the agent's policy Q_t to the *agent's identified-optimum policy* $\pi^*_{\mathrm{ag},t}(\cdot \mid s) := \delta_{a^*_{\mathrm{ag},t}(s)}$ (since (A5)'s base-learner guarantee is what the agent achieves on its identified objective). The trajectory-level dynamic regret in (v) compares to the *true* optimum $\tilde{\pi}^*_t(\cdot \mid s) := \delta_{\tilde{a}^*_t(s)}$. Decompose: $V_t^{\tilde{\pi}^*_t}(s_0) - V_t^{Q_t}(s_0) = \big[V_t^{\tilde{\pi}^*_t}(s_0) - V_t^{\pi^*_{\mathrm{ag},t}}(s_0)\big] + \big[V_t^{\pi^*_{\mathrm{ag},t}}(s_0) - V_t^{Q_t}(s_0)\big]$. Steps 1–3 bound the second bracket. The first bracket is the misidentification penalty. Apply the simulation lemma to the two point-mass policies $(\tilde{\pi}^*_t, \pi^*_{\mathrm{ag},t})$: each per-step bracket reduces to $Q_h^{\tilde{\pi}^*_t}(s_h, \tilde{a}^*_t(s_h)) - Q_h^{\tilde{\pi}^*_t}(s_h, a^*_{\mathrm{ag},t}(s_h)) \in [0, V_{\max}] \cdot \mathbf{1}[\tilde{a}^*_t(s_h) \neq a^*_{\mathrm{ag},t}(s_h)]$. Lower bound by optimality of $\tilde{a}^*$; upper bound by the cumulative-Q range with the indicator on the misidentification event. Taking expectations, with the per-state floor $p_{\mathrm{id}} := \min_s p_{\mathrm{id}}(s)$: $\mathbb{E}\big[V_t^{\tilde{\pi}^*_t}(s_0) - V_t^{\pi^*_{\mathrm{ag},t}}(s_0)\big] \leq V_{\max} \cdot N_h \cdot (1 - p_{\mathrm{id}})$. Summing over T rounds: $V_{\max} \cdot N_h \cdot (1 - p_{\mathrm{id}}) \cdot T$. This is the bias term, in *value coordinates* throughout — no detour through the KL coordinate. Key Lemma 4's KL-readout bound is the right tool for conclusion (iii)'s diagnostic computability; for the value-coordinate misidentification penalty here, the indicator decomposition is direct and tighter (the support condition $Q_t \geq q_0$ at the two argmax candidates is *not* required for this term — only conclusion (iii) needs it).

Combining. Adding Steps 1–4: $\mathbb{E}[\mathrm{DynReg}(T)] \leq 2c\, V_{\max}\, N_h \sqrt{(B_T+1)\, T} + V_{\max}\, N_h\, (1 - p_{\mathrm{id}})\, T$, which is conclusion (v). □

Cesàro tracking corollary Dividing through by T: $\frac{1}{T} \sum_t (V_t^{\tilde{\pi}^*_t} - V_t^{Q_t}) \leq 2c V_{\max} N_h \sqrt{(B_T+1)/T} + V_{\max} N_h (1 - p_{\mathrm{id}})$. As $T \to \infty$ with $B_T = o(T)$ and $p_{\mathrm{id}} \to 1$ (Regime A), the right side tends to zero — the agent's average value over the trajectory tracks the moving optimum's average value. In Regime B the second term is constant; tracking is only up to a $V_{\max}(1 - p_{\mathrm{id}})$ residual. □

Bandit case sharpening (Remark) In the bandit special case ($N_h = 1$, so $\overline{\mathrm{TV}}_t = \mathbb{E}[1 - e^{-K_t}]$), Thompson sampling and UCB (Appendix D) satisfy (A5) with the *stronger logarithmic* per-block rate. By Lattimore and Szepesvári 2020 [27] Theorem 7.1 (or [? auer-cesa-bianchi-fischer-2002-finitetime]), $\mathbb{E}[N_a(L)] = O(\log L/\Delta_a^2)$ per suboptimal arm a; aggregating over arms gives $\mathbb{E}[1 - e^{-K_t}] = \mathbb{E}[1 - Q_t(a^*)] = O(\log t/(t\, \Delta_{\min}^2))$ per stationary block. Substituting and summing gives cumulative regret $O(V_{\max}(B_T+1)\log(T/(B_T+1))/\Delta_{\min}^2)$ — sharper than the worst-case $\sqrt{(B_T+1)T}$ when $\Delta_{\min}$ is bounded away from zero. The framework's $V_{\max} \cdot \mathrm{TV}$ chain is structurally one factor of $\Delta_{\min}$ looser than direct gap-aware UCB analysis (which gives $O(\log T/\Delta_{\min})$ cumulative regret per block by weighting each suboptimal pull by $\Delta(a)$ rather than $V_{\max}$); this is a feature of the unification, not a bug. For $N_h > 1$ MDPs, UCRL2 and UCBVI provide the natural occupancy-weighted instantiation (Appendix D) with c of order $\tilde{O}(\sqrt{N_h SA})$, lifting cleanly to $\tilde{O}(N_h^2 \sqrt{SA(B_T+1)T})$ — same piecewise-stationary B_T family as restart-on-change variants of [1].

Best-of-Both-Worlds non-stationarity extension (Remark on (A5')) Conclusion (v) above uses the restart-on-change form of (A5). The per-round identity route is *stationarity-agnostic* — Steps 1, 2, 4 above hold per round regardless of cross-round dynamics; only Step 3's block-Cauchy–Schwarz aggregator is piecewise-stationary-specific. Replacing (A5) by

> **(A5')** *MASTER-wrapped base learner.* The base learner is the Wei and Luo 2021 [2] MASTER black-box wrapping of any base algorithm satisfying their Assumption 1 (a near-stationary regret guarantee with $C(t) = c_1 t^p + c_2$ for $p \in [1/2, 1)$, plus an auxiliary-quantity output condition; UCRL2, UCBVI, and Thompson sampling all qualify), giving stationary regret $\sum_{t \in W} \mathbb{E}[\overline{\mathrm{TV}}_t] \le 2c\sqrt{|W|}$ within each MASTER-selected window W.

substitutes Step 3's block-Cauchy–Schwarz aggregator with MASTER's adaptive-window aggregator (parallel base-learner instances at exponentially-spaced window sizes $W_k = 2^k$, online meta-learner over the $\log T$ instances). The wrapping is mechanical: the per-round identity (Steps 1–2) is preserved unchanged at each round; the misidentification penalty (Step 4) is round-additive and rides through; only the aggregator changes. The resulting rate, by Wei-Luo Theorem 2 applied with our framework's $C(t) = 2cV_{\max} N_h \sqrt{t}$ at the boundary $p = 1/2$, is $\mathbb{E}[\mathrm{DynReg}(T)] \le \tilde{O}\Big(V_{\max}\, N_h \cdot \min\Big\{\sqrt{(B_T+1)T},\ (V_T+1)^{1/3}T^{2/3}\Big\}\Big) + V_{\max}\, N_h\, (1 - p_{\mathrm{id}})\, T$, *automatically* — without prior knowledge of B_T or V_T. The $V_T^{1/3}T^{2/3}$ exponent matches [3]'s near-optimal RestartQ-UCB rate and is information-theoretically near-optimal: the [30] lower bound $\Omega(K^{1/3}V_T^{1/3}T^{2/3})$ for non-stationary stochastic bandits applies *also at the deterministic-π corner** (their construction is a Bernoulli MAB with dynamic oracle playing the per-round argmax — deterministic-per-round optimum, identical to our canonical scope), so the per-round identity sharpens *constants and per-round form*, not the V_T exponent. The two regimes are *not* structurally distinct: they share the per-round coordinate $1 - e^{-K_t}$ and differ only in the aggregation strategy. □

A.6 Component-by-component failure-mode analysis

This composition is a *bundle of compatible guarantees* (each conclusion supported by its own assumptions), not a single integrated theorem in which every component is necessary for the rate. The rate (v) goes through (A5) and (i) alone; the joint properties — rate plus persistence plus learnability plus corrective-action routing — require all four. The failure modes component-by-component:

Without Component 2 (point-mass identity) The per-round bound is Pinsker $V_{\max}\sqrt{D_{\mathrm{KL}}/2}$ or BH $V_{\max}\sqrt{1 - e^{-D_{\mathrm{KL}}}}$ — both strictly looser at this corner; Pinsker is vacuous for $D_{\mathrm{KL}} > 2$. The metric layer loses *exactness*, and the "behavior-cloning loss against optimal trajectory" interpretation of the cumulative coordinate vanishes.

Without Component 3 (strategic tempo) The agent has no persistence guarantee. By the structural-class theorem on $\mathcal{A}_{\mathrm{decay}}$ (Appendix C.6), every count-accumulating update without forgetting eventually violates the threshold for any positive disturbance; the modeled mismatch $\|\boldsymbol{\delta}_\Sigma\|$ grows unboundedly. The per-round bound (i) still holds, but $K_t(s)$ may itself drift — there is no guarantee the per-round coordinate stays bounded.

Without Component 4 (loop-Level-2) The bound's RHS is computable from Q_t, but its interpretation as regret-against-true-optimum requires identifying $a_t^*(s)$ — an interventional query under the environment's transition kernel. In Regime C ($\iota \approx 0$), the misidentification penalty saturates at $V_{\max}$ per state per step, contributing $V_{\max} N_h \cdot T$ to cumulative regret — the rate (v) becomes vacuous.

Without Component 1 (two-gap diagnostic) The regret signal δ_{regret} alone does not distinguish "policy revision is the right move" from "the goal is unattainable from M_t." An agent might persist in policy revision when the model, policy class, or horizon is the actual problem (Capability-limit cell), or revise the goal when policy revision would have sufficed.

So while the rate (v) goes through (A5) and (i) alone, the *bundle* — rate plus persistence plus learnability plus corrective-action routing — requires all four. Each drop is a concrete loss of an epistemic property that the field treats as separate.

B Proofs of the Key Lemmas

This appendix gives full proofs of the four key lemmas surfaced in Section 5, plus the perturbative extension and the ProST sector-level reduction.

B.1 Proof of Key Lemma 1 (Point-mass reverse-KL/TV identity)

For deterministic $\pi^* = \delta_{a^*}$ and any policy Q (extending $D_{\text{KL}} = +\infty$ when $Q(a^*) = 0$, which gives $1 - e^{-D_{\text{KL}}} = 1 = \text{TV}$ trivially):

The reverse-KL collapses under the point-mass: $D_{\text{KL}}(\delta_{a^*} \,\|\, Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)} = \log \frac{1}{Q(a^*)} = -\log Q(a^*)$, where the convention $0 \log 0 = 0$ disposes of the $a \neq a^*$ terms. Combined with the textbook total-variation calculation $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2}\sum_a |\delta_{a^*}(a) - Q(a)| = \frac{1}{2}\big(|1 - Q(a^*)| + \sum_{a \neq a^*} Q(a)\big) = \frac{1}{2}\big((1 - Q(a^*)) + (1 - Q(a^*))\big) = 1 - Q(a^*)$, we have $-\log Q(a^*) = -\log(1 - \text{TV})$. Solving for TV gives $\text{TV} = 1 - e^{-D_{\text{KL}}}$. $\square$

Strict-improvement substitution Substituting the identity into the Bretagnolle–Huber inequality $\text{TV}(P, Q) \le \sqrt{1 - e^{-D_{\text{KL}}(P \,\|\, Q)}}$ at $P = \delta_{a^*}$ yields $1 - e^{-D_{\text{KL}}} \le \sqrt{1 - e^{-D_{\text{KL}}}}$, which holds *strictly* on $D_{\text{KL}} > 0$ since $x < \sqrt{x}$ on $(0, 1)$. The identity therefore supplies the *exact* TV value at the deterministic-π* corner and lies strictly below the BH envelope there.

Two-sided regret bound (stated as Component 2's two-sided characterization in Section 4.1 and as Key Lemma 1's *Per-round regret bound* paragraph in Section 5.1). From the regret expression $R(Q) = \sum_{a \neq a^*} Q(a)\Delta(a)$ and $\Delta(a) \le V_{\max}$: $R(Q) \le V_{\max} \sum_{a \neq a^*} Q(a) = V_{\max}\big(1 - Q(a^*)\big) = V_{\max}\, \text{TV}(\delta_{a^*}, Q) = V_{\max}(1 - e^{-D_{\text{KL}}})$. The matching action-gap lower bound (full statement in Appendix C.4) gives $R(Q) = \sum_{a \neq a^*} Q(a)\Delta(a) \ge \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min}\, \text{TV}(\delta_{a^*}, Q) = \Delta_{\min}(1 - e^{-D_{\text{KL}}})$. Combining gives the two-sided characterization. The bound is *Lipschitz-equivalent* with constants $\Delta_{\min}/V_{\max}$ and 1, both achieved on extremal value landscapes — therefore *coordinate-optimal among bounds depending only on TV*. $\square$

B.2 Proof of the perturbative extension — ε-stochastic and softmax-regularized

We establish the perturbative identity via a uniform perturbation argument rather than an alignment-specific calculation.

ε-greedy stochastic optimum Let $\pi^*_\epsilon(a^*) = 1 - \epsilon$ and $\pi^*_\epsilon(a) = \epsilon/(|\mathcal{A}| - 1)$ for $a \neq a^*$, and assume the full-support lower bound $Q(a) \ge q_0 > 0$ for all $a \in \mathcal{A}$. (Full support is required since π^*_ϵ has positive mass on every action; without $Q(a) > 0$ on the full support, $D_{\text{KL}}(\pi^*_\epsilon \,\|\, Q) = +\infty$.)

Step 1 — TV is Lipschitz under the point-mass perturbation. $\text{TV}(\pi^*_\epsilon, \delta_{a^*}) = \epsilon$. By the triangle inequality for total variation, $\big|\text{TV}(\pi^*_\epsilon, Q) - \text{TV}(\delta_{a^*}, Q)\big| \le \text{TV}(\pi^*_\epsilon, \delta_{a^*}) = \epsilon$, uniformly in Q — no alignment hypothesis needed.

Step 2 — Reverse-KL admits a uniform expansion. Direct computation:

$$D_{\text{KL}}(\pi^*_\epsilon \,\|\, Q) - D_{\text{KL}}(\delta_{a^*} \,\|\, Q) = (1 - \epsilon)\log\frac{1 - \epsilon}{Q(a^*)} + \tfrac{\epsilon}{|\mathcal{A}|-1} \sum_{a \neq a^*} \log \frac{\epsilon/(|\mathcal{A}| - 1)}{Q(a)} - \big(-\log Q(a^*)\big)$$
$$= (1 - \epsilon)\log(1 - \epsilon) + \epsilon \log Q(a^*) + \epsilon \log \tfrac{\epsilon}{|\mathcal{A}|-1} - \tfrac{\epsilon}{|\mathcal{A}|-1} \sum_{a \neq a^*} \log Q(a), \tag{944}$$

where the second equality collects the $\log Q(a^*)$ contributions via $-(1-\epsilon)\log Q(a^*) + \log Q(a^*) = \epsilon \log Q(a^*)$ and splits the constant log term off the conditional sum. Expanding $(1 - \epsilon)\log(1 - \epsilon) = -\epsilon + O(\epsilon^2)$ and $\epsilon \log(\epsilon/(|\mathcal{A}| - 1)) = \epsilon \log \epsilon - \epsilon \log(|\mathcal{A}| - 1)$, the leading-order behavior is $D_{\text{KL}}(\pi^*_\epsilon \,\|\, Q) = D_{\text{KL}}(\delta_{a^*} \,\|\, Q) + \epsilon \log \epsilon - \epsilon\big(\log(|\mathcal{A}| - 1) + 1\big) + \epsilon \log Q(a^*) - \frac{\epsilon}{|\mathcal{A}|-1} \sum_{a \neq a^*} \log Q(a) + O(\epsilon^2)$. Under $Q(a) \ge q_0$, each $|\log Q(a)| \le \log(1/q_0)$ is uniformly bounded; the dominant correction is the $\epsilon \log \epsilon$ leading term, giving $\big|D_{\text{KL}}(\pi^*_\epsilon \,\|\, Q) - D_{\text{KL}}(\delta_{a^*} \,\|\, Q)\big| = O\big(\epsilon \log(1/\epsilon) + \epsilon \log(1/q_0)\big) = O(\epsilon \log(1/\epsilon))$ absorbing $\epsilon \log(1/q_0)$ into the leading $\epsilon \log(1/\epsilon)$ term as $\epsilon \to 0$ (constants depend on q_0 and $|\mathcal{A}|$).

Step 3 — Map through $1-e^{-x}$ via Lipschitzness. The map $x \mapsto 1-e^{-x}$ is 1-Lipschitz on $[0,\infty)$, so $\left|(1-e^{-D_{\mathrm{KL}}(\pi^*_\epsilon \,\|\, Q)}) - (1-e^{-D_{\mathrm{KL}}(\delta_{a^*} \,\|\, Q)})\right| = O(\epsilon \log(1/\epsilon))$. Combining Steps 1 and 3 with the unperturbed identity $\mathrm{TV}(\delta_{a^*}, Q) = 1-e^{-D_{\mathrm{KL}}(\delta_{a^*} \,\|\, Q)}$: $\mathrm{TV}(\pi^*_\epsilon, Q) = 1-e^{-D_{\mathrm{KL}}(\pi^*_\epsilon \,\|\, Q)} + O(\epsilon\log(1/\epsilon))$ uniformly in Q over the class $\{Q : Q(a) \ge q_0\}$, with constants depending on q_0 and $|\mathcal{A}|$. This establishes the ε-greedy form of the perturbative identity. □

Softmax-regularized optimum Let $\pi^*_\tau(a) \propto \exp(Q_O(a)/\tau)$. Under isolated optimum with action gap $\Delta_{\min}$, the off-optimum mass concentrates exponentially in $1/\tau$: $1-\pi^*_\tau(a^*) = \frac{\sum_{a\neq a^*}\exp(-(Q_O(a^*)-Q_O(a))/\tau)}{1+\sum_{a\neq a^*}\exp(-(Q_O(a^*)-Q_O(a))/\tau)} \le (|\mathcal{A}|-1)\exp(-\Delta_{\min}/\tau)$. Treating π^*_τ as π^*_ϵ with effective $\epsilon = O(\exp(-\Delta_{\min}/\tau))$, the ε-greedy expansion above applies with $\epsilon\log(1/\epsilon) = O\big((\Delta_{\min}/\tau)\exp(-\Delta_{\min}/\tau)\big) = O\big(\tau^{-1}\exp(-\Delta_{\min}/\tau)\big)$, exponentially small with a polynomial prefactor (equivalently, $O(\exp(-c\Delta_{\min}/\tau))$ for any fixed $c \in (0,1)$ since the exponential dominates the polynomial). □

Two-sided regret bound under perturbation Composing with the TV-regret bounds: $\Delta_{\min}\big(1-e^{-D_{\mathrm{KL}}(\pi^*_\epsilon \,\|\, Q)}\big) - O(\epsilon\log(1/\epsilon)) \le R(Q) \le V_{\max}\big(1-e^{-D_{\mathrm{KL}}(\pi^*_\epsilon \,\|\, Q)}\big) + O(\epsilon\log(1/\epsilon))$.

B.3 Proof of Key Lemma 2 (Multi-factor forgetting prerequisite)

We work in the continuous-time idealization of Model (Σ): $\dot{\boldsymbol{\delta}}_\Sigma = -\mathbf{F}(\boldsymbol{\delta}_\Sigma) + \mathbf{w}$ with $\mathbf{F}(\boldsymbol{\delta}_\Sigma) := (\alpha^{\mathrm{ss}}_{ij}\delta_{ij})_{(i,j)\in E}$ the sector correction and $\mathbf{w}$ the deterministic disturbance with $\|\mathbf{w}\| \le \rho_\Sigma$. The discrete-time form recovers the same scaling up to absorbable constants via an AM–GM expansion of $\|\boldsymbol{\delta}_\Sigma - \mathbf{F} + \mathbf{w}\|^2$ (sketched below).

For the Lyapunov function $V(\boldsymbol{\delta}_\Sigma) := \|\boldsymbol{\delta}_\Sigma\|^2$, $\dot V = 2\,\boldsymbol{\delta}_\Sigma^\top \dot{\boldsymbol{\delta}}_\Sigma = -2\,\boldsymbol{\delta}_\Sigma^\top \mathbf{F}(\boldsymbol{\delta}_\Sigma) + 2\,\boldsymbol{\delta}_\Sigma^\top \mathbf{w}$. The first term is the *sector product*, bounded below by the bottleneck: $\boldsymbol{\delta}_\Sigma^\top \mathbf{F}(\boldsymbol{\delta}_\Sigma) = \sum_{(i,j)} \alpha^{\mathrm{ss}}_{ij}\delta^2_{ij} \ge \min_{(i,j)} \alpha^{\mathrm{ss}}_{ij} \cdot \|\boldsymbol{\delta}_\Sigma\|^2 = \mathcal{T}^{\mathrm{bn,ss}}_\Sigma \cdot \|\boldsymbol{\delta}_\Sigma\|^2$, with the bound tight: setting $\boldsymbol{\delta}_\Sigma$ proportional to $\mathbf{e}_{(i^*,j^*)}$ where $(i^*,j^*) \in \arg\min_{(i,j)} \alpha^{\mathrm{ss}}_{ij}$ saturates it. The cross-term is bounded by Cauchy–Schwarz: $|2\,\boldsymbol{\delta}_\Sigma^\top \mathbf{w}| \le 2\,\|\boldsymbol{\delta}_\Sigma\| \cdot \|\mathbf{w}\| \le 2\,\rho_\Sigma\,\|\boldsymbol{\delta}_\Sigma\|$. Combining: $\dot V \le -2\,\mathcal{T}^{\mathrm{bn,ss}}_\Sigma\,\|\boldsymbol{\delta}_\Sigma\|^2 + 2\,\rho_\Sigma\,\|\boldsymbol{\delta}_\Sigma\| = -2\,\|\boldsymbol{\delta}_\Sigma\| \cdot \big(\mathcal{T}^{\mathrm{bn,ss}}_\Sigma\,\|\boldsymbol{\delta}_\Sigma\| - \rho_\Sigma\big)$. The drift is strictly negative whenever $\|\boldsymbol{\delta}_\Sigma\| > \rho_\Sigma/\mathcal{T}^{\mathrm{bn,ss}}_\Sigma$; the trajectory is ultimately bounded by $R^*_\Sigma = \rho_\Sigma/\mathcal{T}^{\mathrm{bn,ss}}_\Sigma$. So whenever $\mathcal{T}^{\mathrm{bn,ss}}_\Sigma > \rho_\Sigma/R_\Sigma$, we have $R^*_\Sigma \le R_\Sigma$ and the modeled mismatch is ultimately bounded within the strategic reserve. This is the standard Khalil sector-Lyapunov ultimate-boundedness setup with the cross-term retained[46] (Section 9.2; comparison-principle form). □

Discrete-time form. For the discrete update $\boldsymbol{\delta}^+_\Sigma = \boldsymbol{\delta}_\Sigma - \mathbf{F}(\boldsymbol{\delta}_\Sigma) + \mathbf{w}$ with $\alpha^{\mathrm{ss}}_{ij} \in [0,1]$ and AM–GM with parameter $\eta = \mathcal{T}^{\mathrm{bn,ss}}_\Sigma/(2(1-\mathcal{T}^{\mathrm{bn,ss}}_\Sigma))$, the steady-state bound is $\|\boldsymbol{\delta}^*_\Sigma\| \le 2\rho_\Sigma/\mathcal{T}^{\mathrm{bn,ss}}_\Sigma$ — same scaling, factor-of-2 absorbable into the threshold constant.

Mean-square corollary. If instead $\mathbf{w}$ is interpreted as zero-mean stochastic disturbance with $\mathbb{E}[\|\mathbf{w}\|^2] \le \rho^2_\Sigma$, the cross-term $\mathbb{E}[2\,\boldsymbol{\delta}_\Sigma^\top \mathbf{w} \mid \boldsymbol{\delta}_\Sigma] = 0$ vanishes in expectation, and the drift inequality gives mean-square ultimate boundedness $\sqrt{\mathbb{E}[V^*]} \le \rho_\Sigma/\sqrt{2\,\mathcal{T}^{\mathrm{bn,ss}}_\Sigma}$ under threshold $\mathcal{T}^{\mathrm{bn,ss}}_\Sigma > \rho^2_\Sigma/(2R^2_\Sigma)$. The deterministic-disturbance form above is the operative one for Theorem 4.1(A2) since non-stationarity-induced drift is structurally adversarial-aligned (consistently moving mismatch in one direction) rather than zero-mean.

Sharpness When the inequality reverses ($\mathcal{T}^{\mathrm{bn,ss}}_\Sigma \le \rho_\Sigma/R_\Sigma$), the adversarial disturbance $\mathbf{w}^*(t) = \rho_\Sigma\,\boldsymbol{\delta}_\Sigma(t)/\|\boldsymbol{\delta}_\Sigma(t)\|$ saturates Cauchy–Schwarz and gives $d\|\boldsymbol{\delta}_\Sigma\|/dt \ge -\mathcal{T}^{\mathrm{bn,ss}}_\Sigma\,\|\boldsymbol{\delta}_\Sigma\| + \rho_\Sigma$. By the comparison principle, $\|\boldsymbol{\delta}_\Sigma(t)\|$ exceeds any $R < \rho_\Sigma/\mathcal{T}^{\mathrm{bn,ss}}_\Sigma$ in finite time. The bottleneck-element refinement: pick $\mathbf{w}$ and $\boldsymbol{\delta}_\Sigma$ both supported on $(i^*,j^*) \in \arg\min_{(i,j)} \alpha^{\mathrm{ss}}_{ij}$ with $|w_{i^*j^*}| = \rho_\Sigma$ — saturates the sector inequality on the contracting side and aligns with $\mathbf{w}$ on the disturbance side simultaneously. The threshold is *sharp inside the diagonal sector model*. The theorem is silent about non-diagonal correction architectures or stabilization mechanisms outside Model (Σ). □

B.4 Proof of the impulsive ProST sector-level reduction

Within Model (Σ), idealize ProST[8] as an impulsive system: between scheduled update times $\{t_1, \ldots, t_K\} \subset [0, T]$ the agent's policy is held fixed and the strategic mismatch evolves under disturbance budget ρ_Σ alone (continuous-destabilizing in the Lyapunov $V = \|\boldsymbol{\delta}_\Sigma\|^2$); at each update time t_i the policy is revised, contracting the modeled mismatch by per-update impulse gain $\gamma \in (0, 1]$ so $\|\boldsymbol{\delta}_\Sigma(t_i^+)\| \leq (1-\gamma)\|\boldsymbol{\delta}_\Sigma(t_i^-)\|$. Then $V(t_i^+) \leq (1-\gamma)^2 V(t_i^-)$.

This fits the Hespanha–Liberzon–Teel[44] (Theorem 1) framework for impulsive systems with destabilizing continuous evolution and stabilizing impulses (the *reverse-ADT* branch). The continuous-evolution rate is $\lambda_c = \rho_\Sigma / R_\Sigma$ (from linearizing the disturbance contribution to $\dot{V} \approx +\lambda_c V + \gamma_c(\rho_\Sigma)$). The impulse contraction in V is $\mu = (1-\gamma)^2$. The reverse-ADT condition for ultimate boundedness — impulses cannot be too sparse — gives $\Delta_{\max} \cdot \lambda_c < -\ln \mu \iff \Delta_{\max} \cdot \rho_\Sigma / R_\Sigma < -\ln(1-\gamma)^2$, where $\Delta_{\max} := \sup_i (t_{i+1} - t_i)$ is the longest inter-update gap. Under the condition the modeled mismatch is ultimately bounded within $R^*_\Sigma \approx \rho_\Sigma \Delta_{\max} / \gamma$ (leading order in γ).

Under uniform blocks $\Delta_i = T/K$ the condition reduces to $K/T > \rho_\Sigma / (2\gamma R_\Sigma)$ to leading order in γ — recovering ProST's K/T form with the impulse gain made explicit. Under nonuniform schedules the impulsive form is *strictly stronger*: the longest block sets the threshold, not the average. The lemma is rigorous for the modeled impulsive sector dynamics; the per-update impulse gain γ is a domain quantity (depends on within-block sample size and update-time discount) and is left abstract. $\square$

B.5 Proof of Key Lemma 3 (Loop generates interventional samples)

By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes o_{t+1}. We invoke Pearl's do-calculus[28] (Theorem 3.4.1, Rule 2 — *action/observation exchange*): $P(o_{t+1} \mid \mathrm{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$ whenever $(o_{t+1} \perp\!\!\!\perp a_t \mid H_t)_{G_{\underline{a_t}}}$, where $G_{\underline{a_t}}$ denotes the graph G with all incoming arrows to a_t removed. In the singular-trajectory + agent-as-policy graph of Section 3, the only arrows into a_t come from H_t (the agent's policy is $\pi_t(a_t \mid H_t)$); deleting them severs the only confounding path from a_t to o_{t+1}. Whether H_t d-separates a_t from o_{t+1} in $G_{\underline{a_t}}$ is precisely the content of (C2). The identity $P(o_{t+1} \mid \mathrm{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$ holds on the support where $\pi_t(a_t \mid H_t) > 0$, which is automatic since a_t was drawn from π_t. $\square$

Remark on equivalent forms of (C2). The d-separation condition $(o_{t+1} \perp a_t \mid H_t)_{G_{\underline{a_t}}}$ is equivalent to the potential-outcome form $a_t \perp Y^{(a_t)} \mid H_t$ (sequential ignorability in the Robins/Murphy sense), and to the truncated-factorization condition $P(\mathbf{U} \mid a_t, H_t) = P(\mathbf{U} \mid H_t)$ where $\mathbf{U}$ are the latent variables jointly affecting a_t and o_{t+1} — applied to the *prior* on $\mathbf{U}$ in the do-marginalization, not to the conditional on o_{t+1}. The three forms are interchangeable; the d-separation form is the most direct handle for the proof above.

On the role of (C1) and (C3). The identification proof above is load-bearing on (C2) alone: (C1)'s full-support positivity and (C3)'s known-action-mechanism are *automatic* under the paper's architecture — (C1) reduces to realized-action positivity ($\pi_t(a_t \mid H_t) > 0$ tautologically since a_t was drawn from π_t); (C3) reduces to agent-policy-queryability (Q_t is the agent's own policy, internally accessible). Both are stated as named conditions for forward-compatibility with off-policy / counterfactual extensions where they would do real work; for the on-policy identification claim of this lemma, (C2) is the substantive condition.

B.6 Proof of Key Lemma 4 (Bias bound)

On the matching event $\{a^*_{\mathrm{ag},t}(s) = \tilde{a}^*_t(s)\}$, the difference $\hat{D}_t(s) - D^{\mathrm{true}}_t(s)$ vanishes identically (both quantities equal $-\log Q_t$ at the same action). On the complementary misidentification event, the support condition $Q_t(\cdot \mid s) \geq q_0$ at *both* $a^*_{\mathrm{ag},t}(s)$ and $\tilde{a}^*_t(s)$ implies $\log Q_t(a^*_{\mathrm{ag},t}(s) \mid s) \in [\log q_0, 0]$ and $\log Q_t(\tilde{a}^*_t(s) \mid s) \in [\log q_0, 0]$, so their absolute difference is at most $\log(1/q_0)$. Therefore $\left|\hat{D}_t(s) - D^{\mathrm{true}}_t(s)\right| \leq \mathbf{1}[a^*_{\mathrm{ag},t}(s) \neq \tilde{a}^*_t(s)] \cdot \log(1/q_0)$. Taking expectations: $\mathbb{E}[|\hat{D}_t(s) - D^{\mathrm{true}}_t(s)|] \leq (1 - p_{\mathrm{id}}(s)) \log(1/q_0)$ by definition of $p_{\mathrm{id}}(s)$. $\square$

High-probability form With probability $1 - \delta$ over a single round's identification event, $\left|\hat{D}_t(s) - D^{\mathrm{true}}_t(s)\right| \leq \log(1/q_0) \cdot \mathbf{1}[\text{misidevent}]$ where $\Pr[\text{misidevent}] = 1 - p_{\mathrm{id}}(s)$. The bias decomposes cleanly: in Regime A ($p_{\text{id}}(s) \to 1$) it vanishes; in Regime B ($p_{\text{id}}(s) \in (0,1)$) it is controlled by misidentification mass; in Regime C ($p_{\text{id}}(s) \to 0$) it saturates at $\log(1/q_0)$. This proves Key Lemma 4. □

Per-state, per-horizon-step lift (used in Theorem 4.1(v)). The lemma applies pointwise at any visited state s_h along a horizon-N_h trajectory: $\mathbb{E}[|\hat{D}_t(s_h) - D_t^{\text{true}}(s_h)|] \leq (1 - p_{\text{id}}(s_h)) \log(1/q_0)$, with $p_{\text{id}}(s_h)$ the per-state argmax-correctness probability. Aggregating linearly over the horizon and assuming the per-state floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$, the per-round bias contribution to the trajectory-level value gap is at most $N_h(1 - p_{\text{id}}) \log(1/q_0)$, contributing the second term of Theorem 4.1(v).

Remarks - Both $a^*_{\text{ag},t}(s)$ and $\tilde{a}^*_t(s)$ must satisfy $Q_t \geq q_0$ for the $\log(1/q_0)$ ceiling to apply; the support condition is on *both* points, not just the agent's estimate. This is *strictly weaker* than the perturbative identity's full-support condition $Q(a) \geq q_0$ for *all* a (Appendix B.2) — the perturbative identity needs full support because π^*_ϵ has positive mass on every action; the bias bound here needs only two-point support. - This is *not* an estimation-from-samples lemma. If Q_t is unavailable internally and must be estimated from on-policy rollouts, an additional concentration argument (Hoeffding under iid fixed-policy samples, or martingale concentration for adaptive policies) is needed; in the architectures this paper considers, Q_t is the agent's policy and is internally available.

C Auxiliary Mathematical Material

This appendix collects supporting material used by the main results: convention hierarchy details, the admissible-divergence family, direction-forcing argument, action-gap matching lower bound, tied-optimum extensions, the structural-class theorem on gain-decay updates, strategic-tempo topologies, deployment modes, and the chain-rule uniqueness theorem.

C.1 Convention hierarchy: C1, C2, C3

Three named continuation conventions form a hierarchy of increasing diagnostic power and computational cost.

C1 — One-step improvement (canonical default) $\pi_{\text{cont}} = \pi_{\text{current}}$. Each action is evaluated assuming current behavior continues afterward. No fixed-point computation, no global-optimality assumption. Cheapest; weakest diagnostic.

C2 — Receding-horizon (N_r-step replanning) At each future step, re-optimize over a horizon of N_r steps using the model available at that step: $\pi_{\text{RH}}(M_\tau) = \arg\max_\pi V_O(M_\tau, \pi;\, N_r)$ applied at each τ. Captures multi-step recovery: a goal that appears unattainable under frozen continuation may be reachable with replanning. Moderate cost; moderate diagnostic.

C3 — Bellman $\pi_{\text{cont}} = \pi^*$ where $\pi^* = \arg\max_\pi V_O(M_t, \pi; N_h)$. The continuation is the optimal policy — a fixed-point equation. Strongest diagnostic; most expensive.

Proposition C.1 (Monotonicity). *For any model M_t, horizon N_h, and policy class Π:* $A_O^{(1)}(M_t; \Pi, N_h) \;\leq\; A_O^{\text{RH}}(M_t; \Pi, N_r, N_h) \;\leq\; A_O^{\text{B}}(M_t; \Pi, N_h)$.

Proof. C1 freezes continuation at π_{current} (possibly suboptimal); C2 re-optimizes periodically, so $\pi_{\text{RH}} \succeq \pi_{\text{current}}$ at each future step; C3 uses the globally optimal π^*. A weakly better continuation yields a weakly higher expected trajectory value; taking the supremum over the first action preserves the ordering. □

Corollary $\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)}$ (since $\delta_{\text{sat}} = V_O^{\min} - A_O$, higher A_O means lower δ_{sat}). And $\delta_{\text{regret}}^{(1)} \leq \delta_{\text{regret}}^{\text{RH}} \leq \delta_{\text{regret}}^{\text{B}}$ (since $\delta_{\text{regret}} = A_O - V_O(M_t, \pi_{\text{current}}; N_h)$, higher A_O means higher δ_{regret}).

C1 is the most conservative diagnostic (most likely to diagnose "locally unattainable"); C3 is the most accurate (least false "unattainable" diagnoses). The 2×2 diagnostic structure is preserved under all three; only the inferential force varies.

C.2 Admissible-divergence family for the regret bound

The reverse-KL direction is forced by the deterministic-π* regime (Appendix C.3). Within the direction-forced family, multiple smooth f-divergences yield valid regret bounds:

Divergence	Bound on $R(Q)$	Tightness	Finite under det. π^*?
$\mathrm{TV}(\pi^*, Q)$	$V_{\max} \cdot \mathrm{TV}$	Tight (extremal V)	Yes
$D_{\mathrm{KL}}(\pi^* \,\\|\, Q)$ via Pinsker	$V_{\max}\sqrt{D_{\mathrm{KL}}/2}$	Loose by $\sqrt{\cdot}$	Yes
$D_{\mathrm{KL}}(\pi^* \,\\|\, Q)$ via point-mass identity	$V_{\max}(1 - e^{-D_{\mathrm{KL}}})$	Tight (this paper)	Yes
$\chi^2(\pi^* \,\\|\, Q)$ (Le Cam)	$V_{\max} \cdot \frac{1}{2}\sqrt{\chi^2}$	Looser than Pinsker	$\chi^2 = 1/Q(a^*) - 1$
$D_\alpha(\pi^* \,\\|\, Q)$ (Rényi, $\alpha \geq 1$)	Various	Interpolates	Yes for $\alpha \geq 1$
$D_{\mathrm{KL}}(Q \,\\|\, \pi^*)$ (forward-KL)	$+\infty$	Vacuous	No

Reverse-KL is selected uniquely within the direction-forced family by the chain-rule axiom (Appendix C.10). The point-mass identity supplies the *exact* bound on reverse-KL under deterministic π^*; the table reflects different bound shapes on the same divergence (with the BH inequality $V_{\max}\sqrt{1 - e^{-D_{\mathrm{KL}}}}$ as the looser general form to which the identity reduces outside the deterministic-π^* corner).

C.3 Direction-forcing argument

For deterministic $\pi^* = \delta_{a^*}$ and any Q with $Q(a) > 0$ for some $a \neq a^*$: $D_{\mathrm{KL}}(Q \,\|\, \pi^*) = \sum_a Q(a) \log \frac{Q(a)}{\pi^*(a)} = \sum_{a \neq a^*} Q(a) \log \frac{Q(a)}{0} = +\infty$. A bound "$R \leq +\infty$" is vacuous. The reverse direction $D_{\mathrm{KL}}(\pi^* \,\|\, Q)$ is finite (and equal to $-\log Q(a^*)$) whenever $Q(a^*) > 0$. The asymmetry is forced by the regime: regret is a one-sided quantity (contributes only from Q's off-optimum mass; π^* has no support off a^*); divergences whose role is to bound this one-sided quantity must themselves be one-sided. Symmetric divergences (squared Hellinger, Jensen-Shannon, symmetrized KL) introduce a vacuous symmetric term.

C.4 Action-gap matching lower bound

For any Q with $\Delta_{\min} = \min_{a \neq a^*} \Delta(a) > 0$: $R(Q) = \sum_{a \neq a^*} Q(a)\Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min} \cdot (1 - Q(a^*)) = \Delta_{\min} \cdot \mathrm{TV}(\pi^*, Q)$. By the point-mass identity (Lemma 5.1), $\mathrm{TV}(\pi^*, Q) = 1 - e^{-D_{\mathrm{KL}}(\pi^* \,\|\, Q)}$, giving the matching lower bound used in the two-sided characterization.

The lower bound is tight when sub-optimal actions are uniformly bad ($\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$). For typical landscapes the gap between upper and lower bound is $V_{\max} - \Delta_{\min}$, controlled by the *spread* of action gaps.

C.5 Tied-optimum and softmax-smoothed extensions

Tied-optimum If π^* has support on a tied-optimum set $\mathcal{A}^* = \{a : Q_O(a) = Q_O(a^*)\}$ with uniform mass, reverse-KL is finite whenever Q covers $\mathcal{A}^*$. The regret bound extends: $R(Q) = \sum_{a \notin \mathcal{A}^*} Q(a)\Delta(a) \leq V_{\max} \cdot \mathbb{P}_Q(a \notin \mathcal{A}^*)$. A multi-action analog of the point-mass identity holds with $\pi^*(a^*) = 1/|\mathcal{A}^*|$ replacing the point-mass form, yielding $D_{\mathrm{KL}}(\pi^* \,\|\, Q) = -\sum_{a \in \mathcal{A}^*} (1/|\mathcal{A}^*|) \log(|\mathcal{A}^*| Q(a))$, a multi-action analog of $-\log Q(a^*)$. The two-sided bound becomes one-sided in general; the upper bound holds with the modified KL form.

Softmax-smoothed π^* with temperature $\tau \to 0$ Replace $\pi^* = \delta_{a^*}$ with $\pi^*_\tau \propto \exp(Q_O/\tau)$. As $\tau \to 0$, $\pi^*_\tau \to \delta_{a^*}$. For finite τ, the *perturbative* identity (Appendix B.2) applies: $\mathrm{TV}(\pi^*_\tau, Q) = 1 - e^{-D_{\mathrm{KL}}(\pi^*_\tau \,\|\, Q)} + O(\tau^{-1}\exp(-\Delta_{\min}/\tau))$ — exponentially small with polynomial prefactor. The two-sided regret bound transfers with the same correction order. Outside the perturbative regime — for genuinely high-entropy optima — the BH inequality $\mathrm{TV} \leq \sqrt{1 - e^{-D_{\mathrm{KL}}}}$ becomes the relevant general bound; Pinsker also applies, with a one-sided regret form.

C.6 The gain-decay structural class $\mathcal{A}_{\text{decay}}$

The persistence inequality $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma/R_\Sigma$ (Section 5.2) is an *instantaneous* check at the current operating point. Define the structural class by gain-decay rather than sample-counting: $\mathcal{A}_{\text{decay}} := \{\text{updates whose effective sector gain } \alpha_{ij}^{(t)} \to 0 \text{ as } t \to \infty \text{ on every revisable element}\}$.

Theorem C.2 (Universal failure of $\mathcal{A}_{\text{decay}}$)**.** *For any fixed (ρ_Σ, R_Σ) with $\rho_\Sigma > 0$, every agent in $\mathcal{A}_{\text{decay}}$ eventually violates the persistence threshold $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma/R_\Sigma$.*

Proof. At every element (i, j), $\alpha_{ij}^{(t)} \to 0$ by definition of $\mathcal{A}_{\text{decay}}$, so $\min_{(i,j)} \alpha_{ij}^{(t)} \to 0$, so the bottleneck $\mathcal{T}_\Sigma^{\text{bn}} \to 0$ with experience. Once $\mathcal{T}_\Sigma^{\text{bn,ss}} < \rho_\Sigma/R_\Sigma$, the threshold is violated. This holds *regardless of prior calibration*: at any finite calibration level, the gain decays below threshold once enough experience accumulates. □

Scope clarification (per-element pull). The class definition requires $\alpha_{ij}^{(t)} \to 0$ on *every* revisable element — i.e., the agent is being pulled at every element repeatedly. Beta-Bernoulli with $\eta_{\text{edge},ij} = 1/(n_{ij} + 1)$ is in the class only for elements actually being updated; an agent that ignores a subset of elements is outside the class for those elements (their gain remains at initial calibration), but inside it for the elements it does touch. The theorem applies to the touched elements; persistence on the untouched ones reduces to whatever calibration was set initially.

This is a *structural* failure of the class, not a tuning problem. The class includes: - Count-accumulating Bayesian updates without forgetting (e.g., Beta-Bernoulli with $\eta_{\text{edge},ij} = 1/(n_{ij}+1)$ where $n_{ij} \to \infty$). - Bounded-memory schemes with growing memory. - Observation-aggregating schemes without restart. - Gradient-based methods with vanishing step sizes ($\eta_t = c/t^\beta$ for $\beta \in (0, 1]$, the Robbins–Monro regime).

Mechanisms *outside* $\mathcal{A}_{\text{decay}}$ — constant-step-size stochastic approximation, sliding-window updates, bounded-memory learners, block-restart schemes, Kalman filters with bounded process noise — maintain a finite gain ceiling and escape asymptotic decay, but then face a *bidirectional* threshold:

Table 1: Bidirectional thresholds for non-accumulating mechanisms outside $\mathcal{A}_{\text{decay}}$.

Mechanism	$\alpha_\Sigma^{\text{ss}}$	Prerequisite holds iff
Exponential forgetting with λ	$1 - \lambda$	$1 - \lambda > \rho_\Sigma/R_\Sigma$
Constant-step SA with rate η	η	$\eta > \rho_\Sigma/R_\Sigma$
Fixed-window mechanisms (sliding-window size W, bounded-memory size K, block-restart period T_R)	$1/W, 1/K, 1/T_R$	window/memory/period $< R_\Sigma/\rho_\Sigma$

For each row, the threshold is *bidirectional* and sharp within the sector-Lyapunov reduction.

C.7 Strategic-tempo across canonical topologies

The bottleneck strategic tempo $\mathcal{T}_\Sigma^{\text{bn,ss}}$ of Lemma 5.2 is verified — and the resulting per-topology threshold derived — across four canonical topologies under Beta-Bernoulli edge updates with per-element forgetting:

- **(T1) Single edge.** $\mathcal{T}_\Sigma^{\text{bn,ss}} = \nu \cdot \iota \cdot (1 - \lambda)$. All three factors load-bearing; setting any one to zero collapses the bottleneck.
- **(T2) Two-edge AND chain, observable intermediate.** $\mathcal{T}_\Sigma^{\text{bn,ss}} = \min\big(\nu_1(1-\lambda_1),\ \nu_1\theta_1(1-\lambda_2)\big)$. Edge 2's effective observation rate is gated by edge 1's success probability θ_1 — *depth-gated attenuation*. For depth-d chains, the bottleneck is $\min_k \prod_{j<k} \theta_j(1 - \lambda_k)$.
- **(T3) Two-edge AND chain, unobservable intermediate.** Per-edge tempo ill-defined; plan-level bottleneck $\mathcal{T}_{\Sigma,\text{plan}}^{\text{bn,ss}} = \nu(1 - \lambda_\Phi)$ over a single tracked plan-level quantity $\hat{\Phi} = p_1 p_2$.

- **(T4) Two-arm OR node, ε-greedy.** $\mathcal{T}_\Sigma^{\mathrm{bn,ss}} = \min\big((1-\varepsilon)(1-\lambda_1),\ \varepsilon(1-\lambda_2)\big)$. Action selection controls rate allocation; pure greedy ($\varepsilon = 0$) collapses the unexplored arm's bottleneck — *exploration-gated*.

The structural decomposition: AND-chains exhibit *depth-gated* geometric attenuation $\nu_k = \nu \prod_{j<k} \theta_j$; OR-nodes exhibit *exploration-gated* allocation. Mixed AND/OR DAGs interleave both. Each topology surfaces a different factor as the binding bottleneck.

Regime-C contamination is structurally fatal $\iota_{ij} = 0$ at any single element $\Rightarrow \mathcal{T}_\Sigma^{\mathrm{bn,ss}} = 0$. An agent with even one fully-confounded element cannot meet the threshold by forgetting alone.

C.8 Three deployment modes of loop-Level-2

The Pearl-do structure of closed-loop interventional access manifests at distinct layers with semantically distinct interventional mechanisms:

- **Mode 1 — agent-self-intervention.** The agent performs do-actions on its own action space as part of its ordinary adaptive loop. Intervention is on the agent's own action; target is the environment's response. This is Lemma 5.3's primary content.
- **Mode 2 — observer-on-sub-agent.** An observer external to a composite performs do-interventions on one sub-agent's action space; target is another sub-agent's response. Reveals cross-coupling structure that component-marginal observation cannot identify.
- **Mode 3 — observer-on-agent-input** (sketched, future work). An observer intervenes on the agent's observation channel; target is the agent's subsequent policy. Useful for offline architectural-class diagnosis.

Modes share the Pearl-do structure but differ in who intervenes on what. The unification is at the pattern level; the mechanism is semantically distinct per layer.

C.9 Distinction from active inference and causal-RL precursors

Action-perception-loop frameworks — active inference[47,48], control-as-inference[49], cybernetics[50,51] — implicitly use the action-causes-observation observation. Our distinctive moves:

- *Bareinboim-hierarchy connection.* Active inference / cybernetics rest on Bayesian-network (Level 1) generative models; we invoke the causal-hierarchy theorem[29] to position the policy DAG as causal with do-conditioning in Q_O.
- *Regime-indexed identifiability (A/B/C).* Active inference literature treats causal identifiability uniformly within modeling assumptions; we surface the regime split at framework level.
- *Scope honesty.* We distinguish "data generated under intervention" from "cleanly identified do-estimates";[45] documents that the active-inference literature sometimes elides this.

The causal-RL line[13–18] is the direct ancestor for regime-indexed identifiability and on-policy interventional access; all are stationary-MDP. Composition with non-stationarity is, to our knowledge, novel.

C.10 Chain-rule uniqueness of reverse-KL

Theorem C.3 (Chain-rule uniqueness (Hobson 1969; Csiszár 1991))**.** *Let $D_f(P \,\|\, Q) = \sum_x Q(x) f(P(x)/Q(x))$ be a smooth f-divergence with f convex and $f(1) = 0$. The chain rule $D_f(P_{XY} \,\|\, Q_{XY}) = D_f(P_X \,\|\, Q_X) + \mathbb{E}_{P_X}\big[D_f(P_{Y|X} \,\|\, Q_{Y|X})\big]$ holds for all joint distributions if and only if $f(t) = c \cdot t \log t$ for some $c > 0$ — i.e., D_f is reverse-KL up to positive scaling.*

Proof. Take any joint P_{XY}, Q_{XY} on a finite product space. Writing $r_x := P(x)/Q(x)$ and $s_{y|x} := P(y \mid x)/Q(y \mid x)$, the joint ratio factorizes as $P_{XY}(x,y)/Q_{XY}(x,y) = r_x \cdot s_{y|x}$. Expanding both sides of the chain rule: $D_f(P_{XY} \,\|\, Q_{XY}) = \sum_{x,y} Q_{XY}(x,y)\, f(r_x s_{y|x}) = \sum_x Q(x) \sum_y Q(y \mid x)\, f(r_x s_{y|x})$, $D_f(P_X \,\|\, Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \,\|\, Q_{Y|X})] = \sum_x Q(x)\, f(r_x) + \sum_x P(x) \sum_y Q(y \mid$

$x)\, f(s_{y|x})$, the second term using $\mathbb{E}_{P_X} = \sum_x P(x) = \sum_x Q(x) r_x$. Equating and collecting: $\sum_x Q(x)\left[\sum_y Q(y \mid x)\, f(r_x s_{y|x}) - f(r_x) - r_x \sum_y Q(y \mid x)\, f(s_{y|x})\right] = 0$. The chain rule must hold for *all* joints, including those where r_x takes arbitrary positive value at one x and $s_{y|x}$ varies independently — which forces the bracketed expression to vanish pointwise. A two-point reduction (take Y binary with $s_{y|x}$ taking values s and $1/s$ on equal-weight $Q(y \mid x)$) collapses the inner sums and yields the functional equation $f(rs) = f(r) + r\, f(s)$ for all $r, s > 0$. With f convex and $f(1) = 0$, the unique solution is $f(t) = c \cdot t \log t$ for $c > 0$[52] (§4) — this gives $D_f(P \,\|\, Q) = c \sum_x Q(x)(P(x)/Q(x)) \log(P(x)/Q(x)) = c \sum_x P(x) \log(P(x)/Q(x)) = c \cdot D_{\mathrm{KL}}(P \,\|\, Q)$, reverse-KL up to positive scaling. □ □

References Hobson 1969 ("A new theorem of information theory," *J. Stat. Phys.*); Csiszár 1991 ("Why least squares and maximum entropy?" *Annals of Statistics*; Theorem 3 corollary, Theorem 5); Shore-Johnson 1980 ("Axiomatic derivation of the principle of maximum entropy," *IEEE Trans. Info. Theory*, system-independence axiom); Sanov 1957 (large-deviation rate function); Aczél-Daróczy 1975 (functional-equation machinery).

These references give *structurally equivalent reformulations* of the same axiom. The Cauchy functional equation each reduces to is the common content. No known uniqueness route outside the independence-on-sub-problems family exists.

Why other family members fail the chain rule Concrete counterexample for χ^2: take Q_X uniform on $\{x_1, x_2\}$, $P_X = (3/4, 1/4)$, $Q(y|x)$ uniform, $P(y|x) = (3/4, 1/4)$. Direct calculation gives $\chi^2(P_{XY} \,\|\, Q_{XY}) = 9/16$ while $\chi^2(P_X \,\|\, Q_X) + \mathbb{E}_{P_X}[\chi^2(P_{Y|X} \,\|\, Q_{Y|X})] = 1/4 + 1/4 = 8/16$. Non-additive. Rényi-$\alpha$ for $\alpha \neq 1$ fails analogously; squared Hellinger likewise fails.

D Algorithm Sketch and Base-Learner Instantiation

A practical algorithm achieving the joint guarantees of Theorem 4.1 requires four components, each translating one of the assumptions (A1)–(A5) into operational form.

(a) Identity-tight base learner satisfying (A5) In the stationary deterministic-π* bandit case ($N_h = 1$), Thompson sampling[19] (posterior-randomized; pointwise $Q_t(a^*) > 0$) and vanilla deterministic UCB[27] (Dirac-deployed; $Q_t(a^*) \in \{0, 1\}$, with the (A1) extended-real reading of Theorem 4.1) both achieve the per-round coordinate in expectation: $\mathbb{E}\left[1 - e^{-K_t}\right] = \mathbb{E}[1 - Q_t(a^*)] = O\left(\frac{\log t}{t\, \Delta^2_{\min}}\right)$, following from Lattimore and Szepesvári 2020[27] Theorem 7.1 ($\mathbb{E}[N_a(t)] \leq 16 \log(t)/\Delta_a^2 + O(1)$ per suboptimal arm; aggregating over arms with the per-state floor gives the per-round form). This satisfies (A5) of Theorem 4.1 with the *stronger logarithmic* rate, yielding cumulative dynamic regret $O(V_{\max}(B_T + 1) \log(T/(B_T + 1))/\Delta^2_{\min})$. The $V_{\max} \cdot$ TV chain is structurally one factor of $\Delta_{\min}$ looser than direct gap-aware UCB analysis (which gives $O(\log T/\Delta_{\min})$ cumulative regret per block by weighting each suboptimal pull by its own $\Delta(a)$ rather than $V_{\max}$); this is a feature of the unification — the framework's contribution is composition + non-stationarity-handling + identity-coordinate exactness, not a sharper bandit constant. For $N_h > 1$ MDPs, UCRL2[53] and UCBVI[41] give per-block cumulative trajectory-level regret $\tilde{O}(N_h^{3/2}\sqrt{SAL})$ with S, A the state/action sizes; rewriting in the occupancy-weighted coordinate satisfies (A5) with c of order $\tilde{O}(\sqrt{N_h SA})$, lifting cleanly to the $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ cumulative rate — same piecewise-stationary B_T family as restart-on-change variants of[1] (the continuous-variation regime of[3] is structurally distinct; see Section 4 *Comparator regime*). The occupancy-weighted form is the natural object for trajectory-level base learners; the per-state KL/TV identity is preserved underneath. Information-directed sampling[20] requires a different analysis here because $H(\pi^*) = 0$ collapses its information ratio at the deterministic-π* corner; we leave the IDS analysis as future work.

(b) Multi-factor forgetting schedule satisfying (A2) Choose per-element discount rates $\{\lambda_{ij}\}$ such that the bottleneck $\mathcal{T}_\Sigma^{\mathrm{bn,ss}} = \min_{(i,j)} \nu_{ij} \iota_{ij} (1 - \lambda_{ij})$ exceeds ρ_Σ / R_Σ. When ρ_Σ is unknown, the[2] black-box reduction gives an adaptive choice (applied per element if heterogeneous); when ρ_Σ

is known via a variation-budget[1] or change-point detection[37], the choice is direct. As a fail-fast pre-check, $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$ is necessary.

(c) Identifiability check satisfying (A3) Test whether the loop data is in Regime A (full identifiability), Regime B (partial), or Regime C (none). In Regime A, no adjustment is needed. In Regime B, apply[17] (DOVI)-style confounding-bias adjustment. In Regime C, the bound is provable but not realizable; algorithm flags the regime as out-of-scope.

(d) Two-gap diagnostic readout satisfying (A4) Compute δ_{sat} (requires A_O, intractable in general; estimable via the policy-class supremum over recent rounds) and δ_{regret} (requires V_O at current policy; tractable in simulation). Route to corrective action class.

The algorithm is sketch-grade; full instantiation and empirical evaluation are deferred to a follow-up paper.

E Pinsker vs. Point-Mass Identity Numerical Comparison

We compare $V_{\max}(1-e^{-D_{\text{KL}}})$ (point-mass identity bound, exact under deterministic π^* per Lemma 5.1 and the two-sided regret theorem) with $V_{\max}\sqrt{D_{\text{KL}}/2}$ (Pinsker bound, also valid under deterministic π^* but loose), $V_{\max}\sqrt{1-e^{-D_{\text{KL}}}}$ (Bretagnolle–Huber inequality, the general envelope below which the identity sits at this corner), and the trivial envelope $V_{\max}$. Set $V_{\max} = 1$ for normalization.

Table 2: Identity vs. Pinsker vs. BH at $V_{\max} = 1$.

D_{KL}	$1-e^{-D_{\text{KL}}}$ (identity)	$\sqrt{1-e^{-D_{\text{KL}}}}$ (BH)	$\sqrt{D_{\text{KL}}/2}$ (Pinsker)	ratio
0.01	0.00995	0.0998	0.0707	7.10×
0.1	0.0952	0.308	0.224	2.35×
0.5	0.393	0.627	0.500	1.27×
1.0	0.632	0.795	0.707	1.12×
2.0	0.865	0.930	1.000	1.16×
4.0	0.982	0.991	1.414	1.02×
10.0	0.99995	0.99998	2.236	1.00×

The point-mass identity bound is uniformly tighter than Pinsker, by a factor of $7\times$ at $D_{\text{KL}} = 0.01$ and converging to 1 as D_{KL} grows large (where both saturate at $V_{\max}$). It is also uniformly tighter than the BH inequality $\sqrt{1-e^{-D_{\text{KL}}}}$ — the relation $x < \sqrt{x}$ on $(0, 1)$ — although both saturate together at $V_{\max}$ for large D_{KL}. Pinsker, by contrast, hits the trivial envelope $V_{\max} = 1$ at $D_{\text{KL}} = 2$ and exceeds it beyond — vacuous from $D_{\text{KL}} = 4$ onward (where the unclipped form returns 1.414), fully vacuous by $D_{\text{KL}} = 10$.

The matching lower bound $\Delta_{\min}(1-e^{-D_{\text{KL}}})$ has the same shape, scaled by $\Delta_{\min}/V_{\max}$. In the extremal value landscape ($\Delta_{\min} = V_{\max}$), the upper and lower bounds coincide and the regret is *identified exactly* by the KL coordinate. For typical landscapes, regret and KL are Lipschitz-equivalent with constants $\Delta_{\min}/V_{\max}$ and 1.

F Prior-Art Retrieval Summary

We conducted a structured prior-art retrieval (via Undermind) with the goal of identifying any framework composing all four target elements: (1) goal-feasibility-vs-policy-quality two-gap diagnostic; (2) an exact reverse-KL/TV regret identity (point-mass) under deterministic-π*, or — more loosely — Bretagnolle–Huber-style use in regret analysis; (3) strategic-tempo / forgetting prerequisite tying convergence to rate of useful policy revision; (4) closed-loop interventional access making regret bounds learnable from on-policy data.

Result: 63 papers retrieved, abstract-level coverage estimated 75%, no direct anticipation in the retrieved set The landscape splits into four largely separate strands corresponding to our four components; we did not find a published framework composing all four in this retrieval. Retrieval queries used the structured prior-art tool Undermind; the four-strand cite-and-distinguish in Section 2 surfaces the closest neighbor in each strand.

Strongest negative signal (in the retrieved corpus): the Bretagnolle–Huber inequality is absent as an upper-bound coordinate for regret analysis in the retrieved RL / non-stationary literature The information-theoretic regret line —[19–24] — uses entropy / mutual information / information ratio / Pinsker / Hellinger; we did not find the BH inequality deployed *as an upper-bound regret coordinate*. (BH is well-known in bandit *lower-bound* proofs via Le Cam's method and Tsybakov's framework [26], including in standard RL textbook treatments [27]; that use is not contested here.) The *exact point-mass identity at the deterministic-π* corner* (Lemma 5.1) — which strictly improves both Pinsker and the BH envelope at this corner — was likewise not found in the retrieved corpus, in either upper-bound or lower-bound use. We frame this as a literature-search observation against an explicitly bounded retrieval (63 papers, ~75% abstract-level coverage), not a formal absence claim against all published RL work; the claim is best read as "to our knowledge / in the retrieved set" rather than "absolutely novel."

Closest neighbors per strand: [1] (Strand 1 — variation-budget dynamic regret); [5] (Strand 2 — closest two-term decomposition, different axis); [8] (Strand 3 — closest tempo result, different form); [14] (Strand 4 — closest causal-RL, stationary only).

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